KINETIC ACCELERATION OF NON-EQUILIBRIUM STEADY STATES

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Abstract. We study whether kinetic lifting still accelerates convergence when the target dynamics is away from detailed balance. Our generators have slow–fast form $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$, where $\gamma > 0$ is friction and $\eta \geq 0$ measures the strength of the non-equilibrium forcing.

The key observation is that, in the Dolbeault–Mouhot–Schmeiser hypocoercive framework [9, 10], the slow coercive block carries the spectral factor $\lambda(Q_O)/(\varpi + \lambda(Q_O))$, where Q_O is the reversible slow backbone. Choosing the corrector shift $\varpi_* = \lambda(Q_O)$ keeps this factor order one and restores the kinetic scale.

Under explicit structural hypotheses—microscopic gap, centered slow–fast coupling, flux regularity, perturbation control, and a kernel-compatible slow gap—we obtain a fully constructive lower bound with computable perturbation threshold $\eta^*(\gamma)$. For uniformly controlled families, $\gamma_{\text{opt}} \asymp \sqrt{\lambda(Q_O)}$, the certified decay rate is of order $\sqrt{\lambda(Q_O)}$, and the admissible forcing threshold is of the same order. We also prove a complementary conditional overdamped ceiling in Hilbert norms where the lifted semigroup is contractive, showing that no rate can beat the diffusive scale as $\gamma \to \infty$.

We also give a reverse-adiabatic-elimination recipe and work out two examples: a convex-potential underdamped Langevin family and a boundary-driven quantum model. In both examples the lower bound is verified rigorously; away from equilibrium the ceiling remains conditional, and numerical experiments support the $\sqrt{\lambda(Q_O)}$ law.

1. Introduction

1.1. **Motivation and main question.** Many systems of interest—forced diffusions [19, 28], boundary-driven quantum systems [23], and actively controlled open-system settings such as active reservoir engineering [20]—operate away from detailed balance. Their generators contain a non-reversible component of strength η and converge to a non-equilibrium steady state (NESS). For reversible systems ($\eta = 0$), lifting to a larger phase space can accelerate mixing [7, 12, 2, 25]: at optimal intermediate friction, the rate improves from the diffusive scale to its square root (quadratic speedup). More recently, non-reversible lifting mechanisms beyond the reversible-target setting have also been studied from a different viewpoint in Markov-process theory [11]. In this paper we ask:

> For a system that is η away from equilibrium, under what conditions does the kinetic acceleration (lifting) persist, and how does the convergence rate degrade as a function of η?

1.2. **The gap-adapted shift.** Concretely, we consider generators on a Hilbert space $\mathcal{H}$ of the form

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}, \tag{1}$$

where $\gamma > 0$ is friction, $\mathcal{R} \leq 0$ is self-adjoint with nontrivial kernel $\mathcal{H}_S := \ker(\mathcal{R})$, $\mathcal{V}_{\text{skew}}$ is skew-adjoint, and $\eta\mathcal{V}_{\text{pert}}$ measures departure from equilibrium. In the singular limit $\gamma \to \infty$, adiabatic elimination [22, 31] reduces the dynamics to an effective slow generator

$$\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$$

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acting on $\mathcal{H}_S$, where P_S is the orthogonal projection onto $\mathcal{H}_S$ and $\mathcal{R}^+$ is the Moore–Penrose pseudoinverse. Its decay scale is $\lambda(\mathcal{L}_O)/\gamma$, so any genuine acceleration must come from finite friction.

The key observation of the paper is that, in the Dolbeault–Mouhot–Schmeiser (DMS) hypocoercive argument, the lower bound is controlled by a 2×2 dissipation matrix. Its slow coercive entry contains the spectral factor

$$\frac{\lambda}{\varpi + \lambda},$$

where $\lambda := \lambda(Q_O)$ is the gap of the reversible slow backbone

$$Q_O := -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S, \tag{2}$$

and $\varpi > 0$ is the shift used in the DMS corrector. If ϖ is kept fixed while $\lambda \downarrow 0$, then the factor is order λ and the certified rate becomes diffusive. If instead one chooses ϖ on the scale of λ, the factor is order one. We take the clean choice

$$\varpi_* := \lambda(Q_O).$$

This single adjustment restores the kinetic square-root law and is the source of every explicit rate in the paper.

1.3. **Main results.** The paper has three outputs, with different logical status:

- **Theorem A**: a fully constructive lower bound, valid once the structural constants are verified;
- **Theorem B**: a *conditional* overdamped ceiling, proving sharpness in Hilbert norms where the lifted semigroup is contractive;
- **Section 6.1**: a reverse-adiabatic-elimination recipe for constructing lifted generators from a prescribed slow model.

Theorem A (Theorem 5.1). *Under Assumptions 4.1–4.2 (and Assumption 4.3 in the unbounded setting), and for perturbations $\eta \le \eta^*(\gamma)$, the semigroup decays exponentially on $\mathcal{H}_0$:*

$$\|e^{t\mathcal{L}} f\| \le C_*(\gamma) e^{-\lambda_\eta t} \|f\|, \qquad f \in \mathcal{H}_0,\ t \ge 0,$$

with explicit lower bound

$$\lambda_\eta \ge (2 - \sqrt{2})\big(\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)\big).$$

The gap-adapted choice $\varpi_* := \lambda(Q_O)$ is the step that restores the kinetic scale. For uniformly controlled families, $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$, the maximal certified rate is of order $\sqrt{\lambda(Q_O)}$, and the threshold $\eta^*(\gamma_{\mathrm{opt}})$ is of the same order.

Theorem B (conditional ceiling; Theorem 5.4). *Assume in addition that the lifted semigroup is contractive in the chosen Hilbert norm. Then any admissible exponential rate ν with prefactor C obeys*

$$\nu \le (1 + \log C)\left[\frac{(1+\kappa_{\mathrm{ell}})\lambda(Q_O)}{\gamma} + \eta(b_1 + b_2) + \eta(a_1 + a_2)\frac{\sqrt{\lambda(Q_O)}}{\gamma}\right].$$

Hence, at fixed η and large γ, the rate cannot beat the diffusive scale $\mathcal{O}(\gamma^{-1}) + \mathcal{O}(\eta)$. In particular, acceleration is a finite-friction effect, and the exponent $1/2$ is optimal whenever the contraction hypothesis holds.

The distinction between Theorems A and B is important. Theorem A is the main certified theorem of the paper: it is constructive and applies model by model once the structural constants are verified. Theorem B is a sharpness statement in the chosen Hilbert norm, but it requires an extra contraction hypothesis. At equilibrium ($\eta = 0$) that hypothesis is automatic; away from equilibrium it is not verified for the nonequilibrium examples treated later, so the ceiling should be read as a conditional complement rather than as part of the model-wise theorem package.

Section 6.1 adds a constructive reverse-adiabatic-elimination recipe: given a target slow generator $\mathcal{L}_O$, we build a lifted generator whose reversible backbone is Q_O. This produces an algorithmic lift

and clarifies steady-state bias. In general $\|P_S h_\eta - g_\eta\| = \mathcal{O}(\eta/\gamma)$, while for designed lifts with $P_F \mathcal{V} P_F = 0$ the bias disappears exactly (Proposition 6.1).

The verified examples are the confining Langevin model on $\mathbb{R}^d$ and a finite-dimensional boundary-driven quantum model.

1.4. **Proof strategy and how to read the paper.**

Strategy of proof. The proof uses the Lyapunov functional $\mathcal{E} = \frac{1}{2}\|f\|^2 + \alpha\,\Re\langle \mathcal{A}_\varpi f, f\rangle$, where $\mathcal{A}_\varpi$ is a shifted resolvent of Q_O. Differentiation yields a 2×2 dissipation matrix for the fast energy and the slow norm. The key point is that $\varpi = \lambda(Q_O)$ makes the slow coercive block order one; the remaining estimates are structural bounds on fast dissipation, transport, and perturbation terms. Optimizing in α and γ then gives the lower bound.

How to read the paper. A first reading can focus on the warm-up in Section 2 and on Theorem 5.1. Sections 3 and 4 supply the abstract setup, Section 6 treats the construction and examples, and the appendices contain the full matrix bookkeeping and model verifications.

1.5. **Related work.**

The L^2 hypocoercive framework based on modified Lyapunov functionals goes back to Dolbeault–Mouhot–Schmeiser [9, 10], building on Villani's H^1 viewpoint [32]. Its Langevin-oriented resolvent correctors were developed by Iacobucci–Olla–Stoltz [19], Roussel–Stoltz [30], and Cao–Lu–Wang [3]. Our contribution stays within this DMS line, but changes the shift: $\varpi_* = \lambda(Q_O)$ replaces the classical fixed shift and upgrades the lower-bound scale from $\mathcal{O}(\lambda^{5/2})$ to $\mathcal{O}(\sqrt{\lambda})$.

The lifting viewpoint originates with Chen–Lovász–Pak [7] and has since been developed by Eberle–Lörler [12, 13], Brigati–Lörler–Wang [2], Li–Lu [25], and Eberle [11]. Away from equilibrium, Dietert [8] and Monmarché [27] provide complementary hypocoercive and ergodic frameworks. Our emphasis is different: an operator-theoretic slow–fast perturbation theory centered on the reversible backbone Q_O, with explicit dependence on the forcing amplitude η and a constructive reverse-adiabatic-elimination recipe. For boundary-driven quantum systems, see [5, 4, 14].

1.6. **Outline.**

Section 2 isolates the mechanism, Sections 3–5 develop the abstract theory, Section 6 treats the construction and the examples, and the appendices contain the functional-analytic details and model verifications.

2. Motivating example: equilibrium Langevin and the gap-adapted DMS corrector

This section gives the equilibrium underdamped Langevin warm-up behind the whole paper. It is the model underlying our later nonequilibrium example, and it isolates the same gap-adapted mechanism with much less bookkeeping: once the DMS corrector is shifted by the overdamped gap m, the slow coercive block becomes order one and the proof reduces to an explicit 2×2 matrix estimate.

For simplicity we normalize the inverse temperature to 1 in this warm-up. Let

$$d\rho_\infty(x,v) = d\mu(x)\,d\kappa(v), \qquad d\mu(x) = Z^{-1}e^{-U(x)}\,dx, \qquad d\kappa(v) = (2\pi)^{-d/2}e^{-|v|^2/2}\,dv,$$

and assume that μ satisfies the Poincaré inequality

$$\|g\|^2_{L^2(\mu)} \le \frac{1}{m}\,\|\nabla_x g\|^2_{L^2(\mu)}, \qquad g \in H^1(\mu),\ \int g\,d\mu = 0, \tag{3}$$

for some $m > 0$. Consider on $L^2_0(\rho_\infty)$ the equilibrium Langevin generator

$$\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma\mathcal{R}, \qquad \mathcal{V}_{\text{skew}} = v\cdot\nabla_x - \nabla_x U\cdot\nabla_v, \qquad \mathcal{R} = -v\cdot\nabla_v + \Delta_v. \tag{4}$$

The slow projection is the velocity average

$$P_S f(x) := \int_{\mathbb{R}^d} f(x,v)\,d\kappa(v), \qquad P_F := \mathrm{id} - P_S,$$

so the slow space consists of functions of x alone. The associated positive overdamped operator is

$$\mathcal{L}_{\mathrm{ovd}} := -Q_O = -\Delta_x + \nabla_x U \cdot \nabla_x, \tag{5}$$

acting on $L^2_0(\mu)$. By (3), $\mathcal{L}_{\mathrm{ovd}} \geq m$ on $L^2_0(\mu)$, equivalently $\lambda(Q_O) = m$.

We use the standard DMS closure argument for this model; the required confining kinetic Fokker–Planck regularity is classical [17, 10, Introduction and §§1–2].

The gap-adapted corrector is

$$\mathcal{A}_m := (m + \mathcal{L}_{\mathrm{ovd}})^{-1} (\mathcal{V}_{\mathrm{skew}} P_S)^\dagger, \tag{6}$$

and the modified Lyapunov functional is

$$\mathsf{E}_\varepsilon(f) := \frac{1}{2}\|f\|^2 - \varepsilon\, \Re\langle \mathcal{A}_m f, f\rangle. \tag{7}$$

Lemma 2.1 (Corrector estimates for the Langevin warm-up)**.** *For every $\phi \in L^2_0(\rho_\infty)$, the corrector $\mathcal{A}_m$ satisfies*

$$\|\mathcal{A}_m \phi\| \leq \frac{1}{2\sqrt{m}}\, \|\phi\|, \tag{8}$$

$$\|\mathcal{V}_{\mathrm{skew}} \mathcal{A}_m \phi\| \leq \|\phi\|, \tag{9}$$

$$\|\mathcal{A}_m \mathcal{V}_{\mathrm{skew}} P_F \phi\| \leq \sqrt{2}\, \|\phi\|. \tag{10}$$

Consequently, if $\varepsilon \leq \sqrt{m}$, then

$$\frac{1 - \varepsilon/\sqrt{m}}{2}\|\phi\|^2 \leq \mathsf{E}_\varepsilon(\phi) \leq \frac{1 + \varepsilon/\sqrt{m}}{2}\|\phi\|^2. \tag{11}$$

Proof. Let $u = \mathcal{A}_m \phi$ and $w = (\mathcal{V}_{\mathrm{skew}} P_S)^\dagger \phi$, so $u = (m + \mathcal{L}_{\mathrm{ovd}})^{-1} w$ and $w \in L^2_0(\mu)$. Since u depends only on x,

$$\|\mathcal{V}_{\mathrm{skew}} u\|^2 = \|\nabla_x u\|^2_{L^2(\mu)} = \langle \mathcal{L}_{\mathrm{ovd}} u, u\rangle_{L^2(\mu)}.$$

Taking the $L^2(\mu)$ inner product of $(m + \mathcal{L}_{\mathrm{ovd}})u = w$ with u and applying Cauchy–Schwarz yields

$$m\|u\|^2 + \|\mathcal{L}^{1/2}_{\mathrm{ovd}} u\|^2 \leq \|\phi\|\, \|\mathcal{L}^{1/2}_{\mathrm{ovd}} u\|.$$

This gives both (8) and (9). For (10), set $B := \mathcal{A}_m \mathcal{V}_{\mathrm{skew}} P_F$. If $g \in \mathrm{Ran}(P_S)$ and $h = (m + \mathcal{L}_{\mathrm{ovd}})^{-1} g$, then an integration by parts shows

$$B^\dagger g = -\sum_{i,j} (v_i v_j - \delta_{ij})\, \partial_{ij} h.$$

The Gaussian fourth-moment identity gives $\|B^\dagger g\|^2 = 2\|\nabla^2_x h\|^2_{L^2(\mu)}$, and the convex Bochner estimate implies $\|\nabla^2_x h\|_{L^2(\mu)} \leq \|\mathcal{L}_{\mathrm{ovd}} h\|_{L^2(\mu)} \leq \|g\|_{L^2(\mu)}$. Hence $\|B^\dagger g\| \leq \sqrt{2}\|g\|$, which is (10). The norm equivalence follows from (8) and Cauchy–Schwarz. □

Write $f_s = P_S f$ and $f_f = P_F f$. The Gaussian Poincaré inequality in v gives $\|f_f\|^2 \leq \|\nabla_v f\|^2$. Therefore, if we set

$$X := \|\nabla_v f\|, \qquad Y := \|f_s\|, \qquad Z := \frac{\gamma}{2\sqrt{m}} + \sqrt{2},$$

then differentiating E_ε along $\partial_t f = \mathcal{L} f$ and using Lemma 2.1 yields

$$\frac{d}{dt}\mathsf{E}_\varepsilon(f) \leq -\begin{pmatrix} X \\ Y \end{pmatrix}^T \begin{pmatrix} \gamma - \varepsilon & -\dfrac{\varepsilon Z}{2} \\ -\dfrac{\varepsilon Z}{2} & \dfrac{\varepsilon}{2} \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}. \tag{12}$$

Indeed, the microscopic dissipation gives $-\gamma X^2$; the macroscopic term is

$$-\varepsilon\Big\langle \frac{\mathcal{L}_{\mathrm{ovd}}}{m+\mathcal{L}_{\mathrm{ovd}}} f_s, f_s\Big\rangle \le -\frac{\varepsilon}{2}Y^2$$

because $\mathcal{L}_{\mathrm{ovd}} \ge m$ on $L^2_0(\mu)$; the fast–fast remainder contributes at most εX^2 by (9); the dual regularity estimate (10) contributes $\varepsilon\sqrt{2}\,XY$; and the correction involving $\gamma\mathcal{R}$ contributes $\varepsilon\gamma(2\sqrt{m})^{-1}XY$ by (8).

Proposition 2.2. *For the gap-adapted corrector $\mathcal{A}_m$, the DMS lower-rate profile satisfies, for every $\gamma > 0$,*

$$\Lambda^{\mathrm{Lan}}_{\mathrm{hypo}}(\gamma) \ge \frac{\gamma}{4\big((\gamma/(2\sqrt{m})+\sqrt{2})^2+2\big)}. \tag{13}$$

The optimum is attained at $\gamma_{\mathrm{opt}} = 4\sqrt{m}$, yielding

$$\Lambda^{\mathrm{Lan}}_{\mathrm{hypo}}(\gamma_{\mathrm{opt}}) \ge \frac{2-\sqrt{2}}{8}\sqrt{m}. \tag{14}$$

Since $m = \lambda(Q_O)$, this is exactly the kinetic scale $\mathcal{O}(\sqrt{\lambda(Q_O)})$.

Proof. Let $M_{\gamma,\varepsilon}$ denote the dissipation matrix in (12). Its determinant is

$$\det M_{\gamma,\varepsilon} = \frac{\gamma\varepsilon}{2} - \frac{\varepsilon^2(Z^2+2)}{4},$$

so the natural choice is

$$\varepsilon_* = \frac{\gamma}{Z^2+2}.$$

For a positive 2×2 matrix, $\lambda_{\min}(M) \ge \det(M)/\operatorname{tr}(M)$, and here $\operatorname{tr}(M_{\gamma,\varepsilon_*}) = \gamma - \varepsilon_*/2 \le \gamma$. Hence

$$\lambda_{\min}(M_{\gamma,\varepsilon_*}) \ge \frac{\gamma}{4(Z^2+2)}.$$

This gives the stated lower bound. Writing $x = \gamma/(2\sqrt{m})$, the right-hand side becomes

$$\sqrt{m}\,\frac{x}{2\big((x+\sqrt{2})^2+2\big)},$$

and a direct derivative computation shows that it is maximized at $x = 2$, i.e. at $\gamma_{\mathrm{opt}} = 4\sqrt{m}$. Substituting $x = 2$ yields (14). □

For comparison, the classical fixed-shift choice $\varpi = 1$ leaves the slow coercive block of order m and yields only the familiar $m^{5/2}$ lower bound. Thus the warm-up already contains the mechanism behind the abstract theory: the gap-adapted choice $\varpi_* = m = \lambda(Q_O)$ restores the kinetic scale, and the later sections ask how much survives after the perturbation $\eta\mathcal{V}_{\mathrm{pert}}$ is switched on.

3. Preliminaries

We set up the abstract Hilbert-space framework, the analytic toolbox for unbounded operators, and the adiabatic-elimination reduction used throughout.

3.1. **Hilbert-space setup.** Let $\mathcal{H}$ be a separable complex Hilbert space with inner product $\langle\cdot,\cdot\rangle$ (conjugate-linear in the first entry) and norm $\|\cdot\| = \langle\cdot,\cdot\rangle^{1/2}$. The generator $\mathcal{L}$ (possibly unbounded) generates a strongly continuous semigroup $(\mathcal{P}_t)_{t\ge 0}$ on $\mathcal{H}$, with adjoint $\mathcal{L}^\dagger$.

Concrete realizations and scope. In the classical setting, $\mathcal{H} = L^2(\mu)$ for a reference measure μ; in the quantum setting, $\mathcal{H} = L^2(\sigma)$ for a faithful state σ, equipped with the KMS inner product $\langle X, Y\rangle_\sigma := \operatorname{tr}(\sigma^{1/2}X^\dagger\sigma^{1/2}Y)$. The verified examples in this paper are the confining Langevin model on $\mathbb{R}^d$ and a finite-dimensional boundary-driven quantum model. Every unbounded operator that appears below is assumed closed and densely defined; in the finite-dimensional example all operators are bounded.

3.2. **Spectral gaps.** Let Π denote the projection onto the stationary direction and $\mathcal{H}_0 := \ker(\Pi)$ the mean-zero subspace. Write $\mathcal{S} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger) \le 0$ for the symmetric part of the generator.

Definition 3.1 (Gap parameters)**.** On $\mathcal{H}_0$ we define the *coercive gap* $\lambda_S := \inf_{f \in \mathcal{H}_0 \setminus \{0\}} (-\langle f, \mathcal{S} f\rangle / \|f\|^2)$ and the *spectral gap* $\lambda(\mathcal{L}) := \inf\{\Re\mu : \mu \in \mathrm{Spec}(-\mathcal{L}|_{\mathcal{H}_0})\}$.

When $\mathcal{L}$ is self-adjoint (detailed balance), $\lambda(\mathcal{L}) = \lambda_S$. In the non-reversible regime, it may happen that $\lambda_S = 0$ while $\lambda(\mathcal{L}) > 0$: the skew part $\frac{1}{2}(\mathcal{L} - \mathcal{L}^\dagger)$ couples degenerate directions to dissipative ones, producing exponential relaxation by hypocoercivity [32, 9]. The standard hypocoercive estimate asserts $\|\mathcal{P}_t f\| \le C e^{-\lambda t}\|f\|$ for $f \in \mathcal{H}_0$; when $C > 1$ is necessary, the relaxation is *strictly hypocoercive*.

Lemma 3.2 (Coercivity characterization)**.** *Assume* $\ker(\mathcal{L}) = \mathbb{C}\mathbf{1}$*. Then:*

(i) If $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$ *and* $-\mathcal{S}|_{\mathcal{H}_0}$ *has a positive spectral gap (automatic in finite dimensions), then the semigroup is coercive* ($C = 1$)*. Conversely, coercivity implies* $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$*.*

(ii) If $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$*, any exponential decay estimate requires* $C > 1$*.*

3.3. **Slow–fast decomposition and adiabatic elimination.** Consider the generator $\mathcal{L}_\gamma = \gamma\mathcal{R} + \mathcal{V}$ with self-adjoint dissipator $\mathcal{R} \le 0$ and interaction $\mathcal{V}$. Define the *slow* and *fast* subspaces

$$\mathcal{H}_S := \ker(\mathcal{R}), \qquad \mathcal{H}_F := \mathcal{H}_S^\perp,$$

with orthogonal projections P_S, $P_F := \mathrm{id} - P_S$. We assume $\mathcal{R}$ has spectral gap $\lambda_R > 0$ on $\mathcal{H}_F$. Any $f \in \mathcal{H}$ decomposes as $f = f_S + f_F$ with $f_S := P_S f$ (slow/macroscopic) and $f_F := P_F f$ (fast/microscopic). In the quantitative theory below, the stationary direction is generated by a vector $\mathbf{1} \in \mathcal{H}_S$, so Π is the orthogonal projection onto $\mathbb{C}\mathbf{1}$. Because $\mathcal{H}_F \perp \mathcal{H}_S$, this implies

$$\mathcal{H}_F \subset \mathcal{H}_0, \qquad \mathcal{H}_0 = (\mathcal{H}_S \cap \mathcal{H}_0) \oplus \mathcal{H}_F.$$

In particular, if $f \in \mathcal{H}_0$, then both $f_S := P_S f$ and $f_F := P_F f$ remain in $\mathcal{H}_0$.

Definition 3.3 (Centered coupling)**.** The interaction satisfies $P_S \mathcal{V} P_S = 0$.

This centering condition ensures that the effective slow dynamics is second-order in γ^{-1}—generated by slow–fast–slow excursions—rather than driven by an $\mathcal{O}(1)$ tangential drift [28, 4].

Definition 3.4 (Effective generator)**.** Let $\mathcal{R}^+$ denote the Moore–Penrose pseudoinverse of $\mathcal{R}$, bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \le \lambda_R^{-1}$. Under the centered coupling condition, we define

$$\mathcal{L}_O := -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S. \tag{15}$$

The operator $\mathcal{L}_O$ is the effective slow dynamics produced by adiabatic elimination. On the macroscopic timescale $\tau = t/\gamma$, the full evolution converges to $e^{\tau \mathcal{L}_O}$ as $\gamma \to \infty$ [4], so the overdamped regime carries the diffusive γ^{-1} bottleneck.

When $\mathcal{V} = \mathcal{V}_{\mathrm{skew}} + \eta \mathcal{V}_{\mathrm{pert}}$, we impose the stronger componentwise centering conditions $P_S \mathcal{V}_{\mathrm{skew}} P_S = 0$ and $P_S \mathcal{V}_{\mathrm{pert}} P_S = 0$. This yields estimates uniform in η and isolates the reversible backbone Q_O.

3.4. **Analytic toolbox for the slow–fast operators.** The functional-analytic foundations of the slow–fast framework—the spectral-calculus construction of $\mathcal{R}^+$ and the Friedrichs construction of Q_O—are standard and collected in Appendix A for completeness. In finite dimensions every step is trivial. We record only the key consequences used later:

(i) $\mathcal{R}^+$ is bounded and self-adjoint with $\|\mathcal{R}^+\| \le \lambda_R^{-1}$ and $\|(-\mathcal{R}^+)^{1/2}\| \le \lambda_R^{-1/2}$;

(ii) Q_O is non-positive and self-adjoint on $\mathcal{H}_S$, satisfying the form identity $\langle \psi, (-Q_O)\psi\rangle = \langle \mathcal{V}_{\mathrm{skew}}\psi, (-\mathcal{R}^+)\mathcal{V}_{\mathrm{skew}}\psi\rangle$;

(iii) for every $\varpi > 0$, $\|(\varpi - Q_O)^{-1}\| \le \varpi^{-1}$, and $\|(\varpi - Q_O)^{-1}|_{\mathcal{H}_{S,0}}\| \le (\varpi + \lambda(Q_O))^{-1}$ when $\lambda(Q_O) > 0$.

4. The non-equilibrium lifting framework

We now state the structural assumptions used for quantitative hypocoercivity at finite friction. The goal is to isolate the ingredients that transfer dissipation from fast modes to slow modes through the DMS corrector [9, 10, 19]: fast spectral gap, centering, flux regularity, and perturbation bounds. Standing convention on the stationary direction. Throughout Sections 4–6, we fix a unit vector $\mathbf{1} \in \mathcal{H}_S$ such that $\mathcal{L}\mathbf{1} = 0$, and denote by Π the orthogonal projection onto $\mathbb{C}\mathbf{1}$. We assume $\Pi e^{t\mathcal{L}} = e^{t\mathcal{L}}\Pi$ for all $t \geq 0$, so $\mathcal{H}_0 := \ker(\Pi)$ is invariant under the semigroup and is preserved by both P_S and P_F.

4.1. Structural assumptions.

Let $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ be an orthogonal slow–fast decomposition. We study generators depending on friction $\gamma > 0$ and perturbation strength $\eta \geq 0$:

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}. \tag{16}$$

The transport term is decomposed as $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$, and P_S, P_F denote the orthogonal projections onto $\mathcal{H}_S, \mathcal{H}_F$.

Before stating the structural assumptions, we record the slow operator that drives the DMS corrector. The reversible contribution to the effective diffusion is

$$Q_O := -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S. \tag{17}$$

Whenever this composition is well defined, Q_O acts on $\mathcal{H}_S$. From the skew-adjointness of $\mathcal{V}_{\text{skew}}$ and the negativity of $\mathcal{R}^+$, it follows that for any $f \in \mathcal{H}_S$,

$$\langle f, Q_O f\rangle = -\langle f, P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S f\rangle = -\langle \mathcal{V}_{\text{skew}}^\dagger f, \mathcal{R}^+\mathcal{V}_{\text{skew}} f\rangle = \langle \mathcal{V}_{\text{skew}} f, \mathcal{R}^+\mathcal{V}_{\text{skew}} f\rangle \leq 0. \tag{18}$$

Thus Q_O is self-adjoint and dissipative on its natural domain; in unbounded settings Assumption 4.3 supplies the required common-core bookkeeping.

Assumption 4.1 (Structural conditions)**.** The generator has the form $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$. We assume:

(1) **Microscopic spectral gap.** $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$, with kernel $\ker(\mathcal{R}) = \mathcal{H}_S$, and there exists $\lambda_R > 0$ such that

$$-\langle \mathcal{R}f, f\rangle \geq \lambda_R\|f\|^2, \qquad \forall f \in \mathcal{H}_F. \tag{19}$$

(2) **Centered coupling.**

$$P_S\mathcal{V}_{\text{skew}}P_S = 0, \qquad P_S\mathcal{V}_{\text{pert}}P_S = 0. \tag{20}$$

(3) **Fast-to-slow regularity.** For $\psi \in \text{dom}(Q_O) \subset \mathcal{H}_S$, define its *microscopic flux* induced by transport as

$$g_\psi := P_F\mathcal{V}_{\text{skew}}P_S\psi \in \mathcal{H}_F. \tag{21}$$

There exist constants $M_R, \kappa_{\text{ell}} > 0$ such that

$$\|g_\psi\|^2 \leq M_R\,\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle, \qquad \forall\psi \in \text{dom}(Q_O) \subset \mathcal{H}_S. \tag{22}$$

$$\|P_F\mathcal{V}_{\text{skew}}P_F\,(-\mathcal{R}^+)\,g_\psi\| \leq \kappa_{\text{ell}}\|P_S\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\psi\| = \kappa_{\text{ell}}\|(-Q_O)\psi\|. \tag{23}$$

(4) **Structural decomposition of the perturbation.**

$$\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}, \tag{24}$$

where

$$\mathcal{K}_{\text{pert}} := \tfrac{1}{2}(\mathcal{V}_{\text{pert}} - \mathcal{V}_{\text{pert}}^\dagger), \qquad \mathcal{S}_{\text{pert}} := \tfrac{1}{2}(\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}^\dagger),$$

with the adjoint taken in $\mathcal{H}$. Assume:

(a) **Skew perturbation control.** The operator $\mathcal{K}_{\text{pert}}$ is skew-symmetric, i.e., for all $f \in \text{dom}(\mathcal{V}_{\text{pert}})$:

$$\Re\langle \mathcal{K}_{\text{pert}} f, f\rangle = 0. \tag{25}$$

There exist constants $a_1, b_1 \geq 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{K}_{\text{pert}} f\| \leq a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|. \tag{26}$$

(b) **Symmetric perturbation control.** $\mathcal{S}_{\text{pert}}$ is symmetric on $\text{dom}((-\mathcal{R})^{1/2})$, and there exist constants $a_2, b_2 \geq 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{S}_{\text{pert}} f\| \leq a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|. \tag{27}$$

These assumptions separate the three ingredients needed later: fast dissipation, slow–fast transport geometry, and perturbation control.

Assumption 4.2 isolates the kernel-compatible slow gap needed for the DMS coercive term.

Assumption 4.2 (Kernel-compatible slow gap)**.** We assume that the stationary direction is the whole kernel of the reversible backbone on the slow space:

$$\ker(Q_O) = \mathbb{C}\mathbf{1} \subset \mathcal{H}_S.$$

Equivalently, on $\mathcal{H}_{S,0} := \mathcal{H}_S \cap \mathcal{H}_0$,

$$-\langle f, Q_O f\rangle \geq \lambda(Q_O)\|f\|^2, \qquad \forall f \in \mathcal{H}_{S,0}. \tag{28}$$

Hence $(\varpi - Q_O)^{-1}$ is bounded on $\mathcal{H}_S$ for every $\varpi > 0$, and in the unbounded setting we additionally assume that it maps $\mathcal{H}_S$ into $\text{dom}(Q_O)$.

Assumption 4.3 (Analytic Core in Unbounded Settings)**.** This assumption is vacuous in finite dimensions. In unbounded settings, fix a dense subspace

$$\mathcal{C} \subset \text{dom}(\mathcal{L}) \cap \text{dom}((-\mathcal{R})^{1/2})$$

satisfying:

(i) **Common core and invariance.** One has $\mathcal{C} \subset \text{dom}(\mathcal{R}) \cap \text{dom}(\mathcal{V}_{\text{skew}}) \cap \text{dom}(\mathcal{V}_{\text{pert}})$, and $\mathcal{C}$ is invariant under the structural operators:

$$\mathcal{R}(\mathcal{C}) \subset \mathcal{C}, \quad \mathcal{V}_{\text{skew}}(\mathcal{C}) \subset \mathcal{C}, \quad P_S(\mathcal{C}) \subset \mathcal{C}, \quad P_F(\mathcal{C}) \subset \mathcal{C}. \tag{29}$$

Moreover, $\mathcal{C}$ is a core for $\mathcal{R}$ and for $\mathcal{V}_{\text{skew}}$, and $\mathcal{C} \cap \mathcal{H}_S$ is a core for Q_O.

(ii) **Slow-to-fast mapping.** The pseudoinverse–transport composition maps the slow core back into the fast core:

$$\mathcal{R}^+\mathcal{V}_{\text{skew}}(\mathcal{C} \cap \mathcal{H}_S) \ \subset \ \mathcal{C} \cap \mathcal{H}_F.$$

(iii) **Closure of the analytic identities.** The operator compositions defining Q_O, $\mathcal{A}_\varpi$, and the bilinear forms used in the Lyapunov and singular-value estimates are well defined on $\mathcal{C}$ and extend continuously to the relevant energy spaces. For every $\varpi > 0$, $(\varpi - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\text{dom}(Q_O)$.

In finite dimensions Assumption 4.3 is automatic; in the Langevin example it is supplied by the standard Schwartz-core theory summarized in Proposition F.1.

4.2. **The DMS corrector.** The DMS corrector is the abstract version of the warm-up resolvent. It solves the shifted slow-space Poisson problem associated with Q_O and supplies the cross term that restores coercivity at the Lyapunov level.

Lemma 4.4 (Resolvent of the slow backbone)**.** *Let Q_O be as in* (17) *and Assumption 4.2. Then for every shift parameter $\varpi > 0$,*

(i) the operator $(\varpi - Q_O)$ is strictly positive on $\mathcal{H}_S$ with spectrum contained in $[\varpi, \infty)$;
(ii) the resolvent $(\varpi - Q_O)^{-1}$ is a bounded, self-adjoint, positive operator on $\mathcal{H}_S$ with $\|(\varpi - Q_O)^{-1}\| \leq \varpi^{-1}$;
(iii) on the mean-zero subspace $\mathcal{H}_{S,0}$, the sharper bound $\|(\varpi - Q_O)^{-1}|_{\mathcal{H}_{S,0}}\| \leq (\varpi + \lambda(Q_O))^{-1}$ holds;
(iv) $(\varpi - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\mathrm{dom}(Q_O)$ by the resolvent mapping property in Assumption 4.3; this is automatic in finite dimensions.

Proof. Since $-Q_O \geq 0$ is self-adjoint with $\ker(Q_O) = \mathbb{C}\mathbf{1}$ (Assumption 4.2), the spectral theorem gives $\sigma(-Q_O|_{\mathcal{H}_{S,0}}) \subset [\lambda(Q_O), \infty)$ and $\sigma(-Q_O|_{\ker Q_O}) = \{0\}$. Therefore $\sigma(\varpi - Q_O) \subset [\varpi, \varpi + \|Q_O\|]$ in finite dimensions, or $\sigma(\varpi - Q_O) \subset [\varpi, \infty)$ in general. Items (i)–(iii) are direct consequences; item (iv) is exactly the extra resolvent mapping property built into Assumption 4.3. □

Definition 4.5 (DMS Corrector)**.** Let $\varpi > 0$ be a shift parameter, to be optimized in Section 5. We define the **corrector** $\mathcal{A}_\varpi : \mathcal{H} \to \mathcal{H}_S$ by

$$\mathcal{A}_\varpi := (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+) P_F. \tag{30}$$

Because dissipation is degenerate on $\mathcal{H}_S$, the plain energy does not produce a spectral gap. The cross term generated by $\mathcal{A}_\varpi$ restores slow coercivity, with remainders controlled by the structural assumptions.

Lemma 4.6 (Structural Estimates for the DMS Corrector)**.** *Under Assumption 4.1, together with Assumption 4.3 in the unbounded setting, the operator $\mathcal{A}_\varpi$ satisfies the following structural estimates, with constants depending solely on the structural parameters $(M_R, \lambda_R, \kappa_{\mathrm{ell}}, a_1, b_1, a_2, b_2)$ and* not *on ϖ:*

(1) ***Corrector Boundedness:***

$$\|\mathcal{A}_\varpi\| \leq k_{\mathcal{A}} \varpi^{-1/2}, \quad \text{with } k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}. \tag{31}$$

(2) ***Regularity Estimates (Forward and Dual):*** *The corrector acts as a bounded regularizer for the skew-symmetric transport in both directions:*

$$\|\mathcal{V}_{\mathrm{skew}} \mathcal{A}_\varpi\| \leq k_{\mathrm{reg}}, \quad \text{with } k_{\mathrm{reg}} := \sqrt{M_R/\lambda_R}, \tag{32a}$$

$$\|\mathcal{A}_\varpi \mathcal{V}_{\mathrm{skew}} P_F\| \leq k_{\mathrm{dual}}, \quad \text{with } k_{\mathrm{dual}} := \kappa_{\mathrm{ell}}. \tag{32b}$$

The dual estimate (32b) *is a direct consequence of the Macroscopic Flux Regularity (Assumption 4.1.3).*

(3) ***Perturbation Control:*** *The interaction with the non-equilibrium perturbation is controlled in both directions:*

$$\|\mathcal{V}_{\mathrm{pert}}^\dagger \mathcal{A}_\varpi\| \leq k_P(\varpi), \tag{33a}$$

$$\|\mathcal{A}_\varpi \mathcal{V}_{\mathrm{pert}}\| \leq \tilde{k}_P(\varpi). \tag{33b}$$

with control constants

$$k_P(\varpi) := (b_1 + b_2) k_{\mathcal{A}}\, \varpi^{-1/2}, \tag{34}$$

$$\tilde{k}_P(\varpi) := \Big((b_1 + b_2) k_{\mathcal{A}} + \tfrac{a_1 + a_2}{2} \Big) \varpi^{-1/2}. \tag{35}$$

Moreover, the centered perturbation coupling in Assumption 4.1.2 implies

$$P_S \mathcal{V}_{\text{pert}} P_S = 0 \qquad \Longrightarrow \qquad P_S \mathcal{V}_{\text{pert}}^{\dagger} P_S = 0, \tag{36}$$

and therefore

$$\mathcal{V}_{\text{pert}}\, \mathcal{H}_S \subset \mathcal{H}_F, \qquad \mathcal{V}_{\text{pert}}^{\dagger}\, \mathcal{H}_S \subset \mathcal{H}_F. \tag{37}$$

Proof. See Appendix C. □

4.3. **The modified Lyapunov functional.**

Definition 4.7 (Modified Lyapunov functional [9, 10])**.** For any $f \in \mathcal{H}$ and a parameter $\alpha \in (0,1)$, we define the functional:

$$\mathcal{E}_{\alpha,\varpi}(f) := \frac{1}{2}\|f\|^2 + \alpha \Re\langle \mathcal{A}_\varpi f, f\rangle. \tag{38}$$

This functional defines a norm equivalent to the Hilbert norm once $\alpha\|\mathcal{A}_\varpi\|$ is small.

Proposition 4.8 (Norm equivalence)**.** *If the parameter α satisfies $\alpha\|\mathcal{A}_\varpi\| \le 1/2$, then for all $f \in \mathcal{H}$:*

$$\frac{1}{2}(1 - 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2 \le \mathcal{E}_{\alpha,\varpi}(f) \le \frac{1}{2}(1 + 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2. \tag{39}$$

Proof. Immediate from $|\Re\langle \mathcal{A}_\varpi f, f\rangle| \le \|\mathcal{A}_\varpi\|\,\|f\|^2$ and the definition of $\mathcal{E}_{\alpha,\varpi}$. □

5. Convergence bounds for the lifted semigroup

We now derive the constructive lower bound and the complementary conditional ceiling that identify the finite-friction acceleration window.

5.1. **Lower bound via modified hypocoercivity.**

We derive the explicit lower bound by optimizing the Lyapunov parameters (α, ϖ); the full matrix estimates are in Appendix D. We package the structural input into the coupling scale $Z(\gamma)$, the optimized weight $\alpha_*(\gamma)$, the rate profile $\Lambda_{\text{hypo}}(\gamma)$, and the perturbative loss $B_{\text{pert}}(\gamma)$.

Rate functions. Define the structural scalars

$$\Theta := \frac{\sqrt{M_R}}{2}, \qquad K_{\text{geom}} := \kappa_{\text{ell}}, \qquad b_{2,\text{eff}} := b_2 + \frac{a_2}{2}(1 + \lambda_R), \tag{40}$$

where Θ measures the slow–fast flux amplitude and K_{geom} collects the transport geometry.

For each friction $\gamma > 0$, define the *coupling scale*

$$Z(\gamma) := \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}, \tag{41}$$

which measures effective slow–fast interaction strength as a function of γ, and set:

$$\alpha_*(\gamma) := \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}, \tag{42}$$

$$\Lambda_{\text{hypo}}(\gamma) := \frac{\frac{1}{4}\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}, \tag{43}$$

$$B_{\text{pert}}(\gamma) := b_{2,\text{eff}} + \alpha_*(\gamma)\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2}, \tag{44}$$

where $C_{\text{sing}} := 2(b_1 + b_2)\, k_{\mathcal{A}} + (a_1 + a_2)$ is the perturbation interaction constant (see Appendix D for the derivation), with $k_{\mathcal{A}}$ the resolvent norm bound from Lemma 4.6. Here $\alpha_*(\gamma)$ is the optimized Lyapunov weight, $\Lambda_{\text{hypo}}(\gamma)$ is the lower-rate profile, and $B_{\text{pert}}(\gamma)$ is the perturbative loss. The apparent factor $\lambda(Q_O)^{-1/2}$ in B_{pert} is cancelled at optimal friction; see Corollary 5.2.

We also isolate the explicit admissibility restriction coming from the symmetric perturbation in the basic energy estimate:

$$\eta_{\mathrm{sym}}(\gamma) := \begin{cases} +\infty, & a_2 = 0, \\ 2\gamma/a_2, & a_2 > 0. \end{cases} \tag{45}$$

This is exactly the condition under which the coefficient of $\|(-\mathcal{R})^{1/2} f\|^2$ remains nonnegative in the basic energy estimate; see Appendix D.

With the gap-adapted choice $\varpi_* := \lambda(Q_O)$, define the associated Lyapunov-distortion parameter and norm-equivalence prefactor by

$$\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|, \tag{46}$$

$$C_*(\gamma) := \sqrt{\frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}}. \tag{47}$$

The appendix proves the uniform bound $\delta_*(\gamma) \le 2^{-1/2}$ for all $\gamma > 0$. Hence Proposition 4.8 applies to the optimized Lyapunov functional and $C_*(\gamma) \le 1 + \sqrt{2}$.

Theorem 5.1 (Explicit quantitative hypocoercivity)**.** *Suppose Assumptions 4.1 and 4.2 hold, and in the unbounded setting also Assumption 4.3. Set the explicit certified threshold*

$$\eta^*(\gamma) := \min\left\{\eta_{\mathrm{sym}}(\gamma),\ \frac{\Lambda_{\mathrm{hypo}}(\gamma)}{2B_{\mathrm{pert}}(\gamma)}\right\}.$$

For any $\gamma > 0$, if $\eta \le \eta^*(\gamma)$, then $e^{t\mathcal{L}}$ is exponentially stable on the invariant mean-zero subspace $\mathcal{H}_0 = \ker(\Pi)$:

$$\|e^{t\mathcal{L}} f\| \le C_*(\gamma)\, e^{-\lambda_\eta t}\|f\| \le (1+\sqrt{2})\, e^{-\lambda_\eta t}\|f\|, \quad \forall t \ge 0,\ \forall f \in \mathcal{H}_0. \tag{48}$$

The global decay rate λ_η satisfies the strict lower bound:

$$\lambda_\eta \ge \frac{\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)}{1+\delta_*(\gamma)} \ge (2-\sqrt{2})\left(\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)\right). \tag{49}$$

Proof outline. The full computation is in Appendix D. On the analytic core, differentiating $\mathcal{E}_{\alpha,\varpi_*}$ yields a 2×2 dissipation matrix for the fast energy and the slow norm. The gap-adapted choice $\varpi_* = \lambda(Q_O)$ makes the slow coercive entry order one, while the perturbation contributes only $\mathcal{O}(\eta)$ corrections to the fast block and the coupling terms.

Optimizing the determinant of this matrix gives the weight $\alpha_*(\gamma)$ and the profile $\Lambda_{\mathrm{hypo}}(\gamma)$. The solver bound for $\mathcal{A}_{\varpi_*}$ then yields $\delta_*(\gamma) \le 2^{-1/2}$, hence the norm-equivalence constant $C_*(\gamma) \le 1+\sqrt{2}$.

The differential inequality is first proved on the analytic core and then extended to all of $\mathcal{H}_0$ by density and boundedness of $\mathcal{A}_{\varpi_*}$, giving the stated semigroup estimate and rate bound. □

Corollary 5.2 (Kinetic acceleration and singularity cancellation)**.** *Assume the setting of Theorem 5.1. Assume furthermore that the structural constants Θ, K_{geom}, k_{reg}, λ_R, $b_{2,\mathrm{eff}}$, and C_{sing} are independent of $\lambda(Q_O)$ (in particular, they remain $\mathcal{O}(1)$ as $\lambda(Q_O) \downarrow 0$), and that $b_{2,\mathrm{eff}} > 0$.*

Then the unperturbed rate function $\Lambda_{\mathrm{hypo}}(\gamma)$ defined in (43) attains its global maximum at

$$\gamma_{\mathrm{opt}} = \frac{\sqrt{\lambda(Q_O)}}{\Theta}\sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}} \asymp \sqrt{\lambda(Q_O)}. \tag{50}$$

At this friction scale, the maximal lower bound exhibits kinetic scaling:

$$\max_{\gamma>0} \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}. \tag{51}$$

Moreover, the mixing weight in (44) *satisfies*

$$\alpha_*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}, \qquad \alpha_*(\gamma_{\rm opt})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1), \tag{52}$$

so the solver singularity $\lambda(Q_O)^{-1/2}$ *in* $B_{\rm pert}(\gamma)$ *is cancelled at the optimal friction. Consequently,* $B_{\rm pert}(\gamma_{\rm opt}) = \mathcal{O}(1)$ *and the certified threshold satisfies*

$$\eta^*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}. \tag{53}$$

Proof. Write

$$\Lambda_{\rm hypo}(\gamma) = \frac{\lambda_R}{4}\,\frac{\gamma}{\big(\gamma\Theta\lambda(Q_O)^{-1/2} + K_{\rm geom}\big)^2 + 2k_{\rm reg}}. \tag{54}$$

Set $x := \gamma\Theta\lambda(Q_O)^{-1/2}$ and $C_0 := 2k_{\rm reg}$. Then $\Lambda_{\rm hypo}(\gamma) = \frac{\lambda_R}{4\Theta}\sqrt{\lambda(Q_O)}\,\frac{x}{(x+K_{\rm geom})^2+C_0}$. A direct derivative computation shows that the function $x \mapsto \frac{x}{(x+K_{\rm geom})^2+C_0}$ is maximized at $x = \sqrt{K_{\rm geom}^2 + C_0}$, giving the stated $\gamma_{\rm opt}$ and the scaling $\max_\gamma \Lambda_{\rm hypo}(\gamma) \asymp \sqrt{\lambda(Q_O)}$.

For the perturbation term, by (44), $B_{\rm pert}(\gamma) = b_{2,\rm eff} + \alpha_*(\gamma)\,C_{\rm sing}\,\lambda(Q_O)^{-1/2}$. At $\gamma = \gamma_{\rm opt}$ we have $x = \mathcal{O}(1)$ and $Z(\gamma_{\rm opt}) = x + K_{\rm geom} = \mathcal{O}(1)$, hence $\alpha_*(\gamma_{\rm opt}) = \frac{\gamma_{\rm opt}\lambda_R}{Z(\gamma_{\rm opt})^2+2k_{\rm reg}} = \mathcal{O}(\sqrt{\lambda(Q_O)})$, which implies $\alpha_*(\gamma_{\rm opt})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1)$ and therefore $B_{\rm pert}(\gamma_{\rm opt}) = \mathcal{O}(1)$. Finally, the matrix-controlled threshold satisfies $\Lambda_{\rm hypo}(\gamma_{\rm opt})/(2B_{\rm pert}(\gamma_{\rm opt})) \asymp \sqrt{\lambda(Q_O)}$. If $a_2 = 0$, then $\eta_{\rm sym}(\gamma_{\rm opt}) = +\infty$ and therefore $\eta^*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}$. If $a_2 > 0$, then $\eta_{\rm sym}(\gamma_{\rm opt}) = 2\gamma_{\rm opt}/a_2 \asymp \sqrt{\lambda(Q_O)}$. In either case, $\eta^*(\gamma_{\rm opt}) = \min\{\eta_{\rm sym}(\gamma_{\rm opt}), \Lambda_{\rm hypo}(\gamma_{\rm opt})/(2B_{\rm pert}(\gamma_{\rm opt}))\} \asymp \sqrt{\lambda(Q_O)}$. □

For the equilibrium Langevin warm-up of Section 2, the general formulas recover Proposition 2.2 exactly.

5.2. **Conditional overdamped diffusive ceiling.** We now prove the complementary sharpness statement: if the lifted semigroup is contractive in the chosen Hilbert norm, then no L^2 decay rate can beat the diffusive overdamped scale. At equilibrium this extra input is automatic; away from equilibrium it is not verified for the examples below.

Define the *singular-value gap*

$$s(\mathcal{L}) := \inf_{f\in\mathcal{H}_0\cap{\rm dom}(\mathcal{L})\setminus\{0\}} \frac{\|\mathcal{L}f\|}{\|f\|}, \tag{55}$$

that is, the smallest singular value of $\mathcal{L}|_{\mathcal{H}_0}$ in finite dimensions, and the corresponding infimum over the operator domain in unbounded settings.

Assumption 5.3 (Overdamped ceiling hypotheses)**.** In addition to Assumptions 4.1, 4.2 (and 4.3 in the unbounded setting), we assume:

(i) $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$;
(ii) $\lambda(Q_O)$ is attained by a normalized eigenfunction $\phi \in {\rm dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0$ (automatic in finite dimensions and many compact-resolvent settings);
(iii) the first-order corrector $\psi := \mathcal{R}^+\mathcal{V}_{\rm skew}\phi$ belongs to ${\rm dom}(\mathcal{L}) \cap {\rm dom}((-\mathcal{R})^{1/2})$ with $P_F\mathcal{V}_{\rm skew}\psi$ well defined;
(iv) $e^{t\mathcal{L}}$ is a contraction semigroup on $\mathcal{H}_0$.

Theorem 5.4 (Conditional overdamped diffusive ceiling via singular value gap)**.** *Assume Assumption 5.3. Let* $C \geq 1$ *and* $\nu > 0$ *be such that*

$$\|e^{t\mathcal{L}}f\| \leq C\,e^{-\nu t}\,\|f\|, \qquad \forall t \geq 0,\ \forall f \in \mathcal{H}_0. \tag{56}$$

Then for every $\gamma > 0$,

$$\nu \le (1+\log C)\, s(\mathcal{L}) \le (1+\log C)\left[\frac{(1+\kappa_{\mathrm{ell}})\,\lambda(Q_O)}{\gamma} + \eta\,(b_1+b_2) + \eta\,(a_1+a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}\right], \tag{57}$$

where κ_{ell} is the flux-regularity constant from Assumption 4.1.3 and a_1, a_2, b_1, b_2 are from Assumption 4.1.4. In particular, for $\eta \le c/\gamma$ one obtains the diffusive overdamped scaling $\nu = \mathcal{O}(\gamma^{-1})$ as $\gamma \to \infty$.

Remark 5.5 (Norm dependence of the ceiling). Theorem 5.4 is an upper bound in the *chosen* Hilbert norm. Away from equilibrium, passing to a stationary-state-adapted norm would require re-proving contraction and re-establishing the structural constants, so for $\eta \neq 0$ the theorem should be read as a conditional ceiling rather than as a verified model-wise statement for the examples.

Proof outline. The proof is given in Appendix E. Testing $s(\mathcal{L})$ on the first-order corrector $f_\gamma := \phi - \gamma^{-1}\psi$, built from a slow eigenfunction $Q_O\phi = -\lambda(Q_O)\phi$ and $\psi := \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi$, yields the upper bound on $s(\mathcal{L})$. The only nontrivial fast remainder is controlled by the flux-regularity estimate (23), and the perturbation terms are bounded by the $(-\mathcal{R})^{1/2}$-energy estimates from Assumption 4.1.4.

The inequality $\nu \le (1+\log C)\, s(\mathcal{L})$ then follows from Lemma E.1, combining contraction on short times with the assumed exponential decay on long times. □

The theorem shows that in the overdamped regime the best achievable L^2 decay rate is bounded by the diffusive scale $\lambda(Q_O)/\gamma$, up to explicit perturbative terms. Combined with Theorem 5.1 and Corollary 5.2, this shows that the accelerated regime must occur at intermediate friction $\gamma \asymp \sqrt{\lambda(Q_O)}$.

At equilibrium ($\eta = 0$), the contraction hypothesis is automatic because $\mathrm{Re}\langle \mathcal{L}f, f\rangle = \gamma\langle \mathcal{R}f, f\rangle \le 0$. Hence Theorems 5.1 and 5.4 match and show that the exponent 1/2 is optimal. Away from equilibrium, the ceiling remains conditional in the present examples.

6. Examples and verification of the mechanism

This section first turns the abstract slow–fast picture into a concrete reverse-adiabatic-elimination recipe and then verifies the structural hypotheses for two representative families.

Status. Theorem 5.1 applies rigorously in both examples; Theorem 5.4 remains conditional away from equilibrium.

6.1. **Reverse adiabatic elimination: from effective dynamics to lifted dynamics.** Given a non-equilibrium effective generator $\mathcal{L}_O$ on the slow space $\mathcal{H}_S$, we seek a lifted generator $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ whose adiabatic elimination (Definition 3.4) is exactly $\mathcal{L}_O$. Reverse adiabatic elimination has three steps: identify the reversible backbone Q_O, choose a fast space and skew coupling that reproduce it, and then choose the perturbation so that the effective generator is exactly $\mathcal{L}_O$. In finite dimensions this is algebraic; in unbounded settings one must also verify the domain properties.

Perturbation expansion of $\mathcal{L}_O$. By Definition 3.4, $\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$ with $\mathcal{V} = \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$. Expanding in powers of η gives

$$\mathcal{L}_O = \underbrace{-P_S\mathcal{V}_{\mathrm{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{skew}}\,P_S}_{Q_O} \underbrace{-\,\eta\,P_S\big(\mathcal{V}_{\mathrm{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{pert}} \;+\; \mathcal{V}_{\mathrm{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{skew}}\big)P_S}_{\text{cross terms } \mathcal{O}(\eta)} \underbrace{-\,\eta^2\,P_S\mathcal{V}_{\mathrm{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{pert}}\,P_S}_{\text{pure perturbation } \mathcal{O}(\eta^2)}. \tag{58}$$

The first term is the reversible backbone Q_O; the other two are the $\mathcal{O}(\eta)$ and $\mathcal{O}(\eta^2)$ corrections. In the boundary-driven example they are written explicitly in (79).

Step 1: Identify Q_O. Set $\eta = 0$ in the slow model. The result is Q_O, whose spectral gap $\lambda(Q_O)$ is the basic scale in Theorem 5.1.

Step 2: Choose fast space and coupling to match Q_O. The adiabatic-elimination identity implies

$$Q_O = T^\dagger (\mathcal{R}_F)^{-1} T, \qquad T := P_F \mathcal{V}_{\text{skew}} P_S, \tag{59}$$

where $\mathcal{R}_F := \mathcal{R}|_{\mathcal{H}_F} < 0$. So, after choosing $\mathcal{H}_F$ and $\mathcal{R}_F$, we pick $T : \mathcal{H}_S \to \mathcal{H}_F$ to satisfy $T^\dagger \mathcal{R}_F^{-1} T = Q_O$. A universal choice is $\mathcal{H}_F = \mathcal{H}_S$, $\mathcal{R}_F = -\lambda_R \,\mathrm{id}$, $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. Then $T^\dagger \mathcal{R}_F^{-1} T = Q_O$, and we set $\mathcal{R} := 0 \oplus \mathcal{R}_F$ and $\mathcal{V}_{\text{skew}} := \left(\begin{smallmatrix} 0 & -T^\dagger \\ T & 0 \end{smallmatrix}\right)$.

This universal choice is the one used in the algorithmic-lift comparisons below.

Step 3: Determine $\mathcal{V}_{\text{pert}}$ from consistency. With $\mathcal{R}$ and $\mathcal{V}_{\text{skew}}$ fixed, we choose $\mathcal{V}_{\text{pert}}$ so that adiabatic elimination reproduces $\mathcal{L}_O$. This is equivalent to the consistency relation

$$\mathcal{L}_O - Q_O \;=\; -\eta\, P_S\big(\mathcal{V}_{\text{skew}}\, \mathcal{R}^+\, \mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}\, \mathcal{R}^+\, \mathcal{V}_{\text{skew}}\big) P_S \;-\; \eta^2\, P_S \mathcal{V}_{\text{pert}}\, \mathcal{R}^+\, \mathcal{V}_{\text{pert}}\, P_S, \tag{60}$$

with the centering constraint $P_S \mathcal{V}_{\text{pert}} P_S = 0$. In the Langevin example, $\mathcal{V}_{\text{pert}} P_S = 0$, so the formula simplifies strongly; in the boundary-driven example all terms are present. In finite dimensions this is an algebraic problem, usually with non-unique solutions; in infinite dimensions solvability is model-dependent.

Proposition 6.1 (Unbiasedness of designed lifts)**.** *Consider a lift $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ constructed via the reverse adiabatic elimination recipe above, with $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^\dagger \\ T & 0 \end{smallmatrix}\right)$ (Step 2) and $\mathcal{V}_{\text{pert}}$ satisfying the centering condition $P_S \mathcal{V}_{\text{pert}} P_S = 0$ and the consistency condition* (60)*. If additionally $P_F \mathcal{V}_{\text{pert}} P_F = 0$ (the perturbation has no fast-to-fast component), then:*

(i) $P_F \mathcal{V} P_F = 0$;
(ii) the Schur complement (61) *satisfies $\gamma\, S(\gamma) = \mathcal{L}_O$ for all $\gamma > 0$;*
(iii) whenever the lifted and effective stationary states are well defined, the lifted NESS satisfies $P_S h_\eta = g_\eta$ for all $\gamma > 0$, i.e., the projection bias $\Delta_{\text{bias}}(\gamma) \equiv 0$.

The condition $P_F \mathcal{V}_{\text{pert}} P_F = 0$ is a design choice: (60) *constrains only the off-diagonal blocks, so the fast–fast block may be set to zero without changing the effective dynamics.*

Proof. (i) Since $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^\dagger \\ T & 0 \end{smallmatrix}\right)$ is block-off-diagonal, $P_F \mathcal{V}_{\text{skew}} P_F = 0$. Combined with $P_F \mathcal{V}_{\text{pert}} P_F = 0$, we get $P_F \mathcal{V} P_F = P_F(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}) P_F = 0$. (ii) With $P_F \mathcal{V} P_F = 0$, the Schur complement (61) reduces to $S(\gamma) = -P_S \mathcal{V}(\gamma\mathcal{R}_F)^{-1} \mathcal{V}\, P_S = \gamma^{-1}\mathcal{L}_O$. (iii) Since $\gamma S(\gamma) = \mathcal{L}_O$, the NESS equation $S(\gamma)u = 0$ becomes $\mathcal{L}_O u = 0$, so $u = g_\eta$. □

Proposition 6.1 gives the design rule used later: if $P_F \mathcal{V}_{\text{pert}} P_F = 0$, then the projected lifted steady state equals the overdamped one for all γ. In general the slow marginal $u = P_S h_\eta$ satisfies the Schur complement equation

$$S(\gamma)\, u = 0, \qquad S(\gamma) := -P_S \mathcal{V}\big(\gamma\mathcal{R}_F + P_F \mathcal{V} P_F\big)^{-1} \mathcal{V}\, P_S. \tag{61}$$

Under the usual isolated-eigenvalue perturbation hypotheses, this gives $\|P_S h_\eta - g_\eta\| = \mathcal{O}(\eta/\gamma)$. When $P_F \mathcal{V} P_F = 0$, as in Proposition 6.1, the bias vanishes identically. This is the construction used in the numerical algorithmic lifts below.

6.2. **Example 1: Nonequilibrium Langevin family with convex potential.** We study the underdamped Langevin family on $\mathbb{R}^{2d}$ with a uniformly convex confining potential and a bounded non-gradient perturbation. This is our main infinite-dimensional example: it verifies Theorem 5.1 with constants uniform as $m \downarrow 0$, while the ceiling remains conditional away from equilibrium in the present $L^2(\mu)$ norm. Throughout this example we fix $d \geq 2$, $\beta > 0$, $m \in (0, 1]$, and take $U \in C^\infty(\mathbb{R}^d)$ to satisfy

$$\nabla^2 U(q) \geq m \quad \text{for all } q \in \mathbb{R}^d, \tag{62}$$

with all derivatives of positive order growing at most polynomially at infinity. These regularity assumptions are used only to place the model within the standard confining Kramers–Fokker–Planck

framework; they do not affect the explicit constants below. We also assume $F \in C_b^\infty(\mathbb{R}^d;\mathbb{R}^d)$ is bounded and not of gradient form, so $\mathcal{L}_O$ is genuinely nonequilibrium for every $\eta \neq 0$.

Setup. Let

$$d\mu(q,p) = Z^{-1}\exp\Big(-\beta\big(\tfrac{1}{2}|p|^2 + U(q)\big)\Big)\,dq\,dp = d\mu_q(q)\,d\mu_p(p), \tag{63}$$

where $\mu_q(dq) \propto e^{-\beta U(q)}\,dq$, and work in $\mathcal{H} = L^2(\mu)$ with slow projection

$$P_S f = \Pi_p f := \int_{\mathbb{R}^d} f(\cdot,p)\,d\mu_p(p). \tag{64}$$

We use the Schwartz core

$$\mathcal{C}_{\mathrm{Lan}} := \mathcal{S}(\mathbb{R}^d_q \times \mathbb{R}^d_p). \tag{65}$$

Its semigroup and common-core properties are summarized in Proposition F.1.

Generator decomposition. The Langevin generator is written as

$$\mathcal{L} = \mathcal{V}_{\mathrm{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\mathrm{pert}}, \tag{66}$$

with

$$\mathcal{V}_{\mathrm{skew}} := p\cdot\nabla_q - \nabla_q U\cdot\nabla_p, \qquad \mathcal{R} := -p\cdot\nabla_p + \beta^{-1}\Delta_p, \qquad \mathcal{V}_{\mathrm{pert}} f := F(q)\cdot\nabla_p f. \tag{67}$$

Effective generator $\mathcal{L}_O$ (explicit formula). Since $\mathcal{R}(p_i) = -p_i$ and $\int p_i\,d\mu_p = 0$, Definition 3.4 gives, for $\phi = \phi(q)$,

$$\mathcal{L}_O\phi = \beta^{-1}\Delta_q\phi - (\nabla_q U - \eta F(q))\cdot\nabla_q\phi, \tag{68}$$

and therefore

$$Q_O\phi = \beta^{-1}\Delta_q\phi - \nabla_q U\cdot\nabla_q\phi. \tag{69}$$

The slow reference measure is $\mu_q(dq) \propto e^{-\beta U(q)}\,dq$. The Bakry–Émery criterion [1], applied to the uniform convexity $\nabla^2 U \geq m > 0$, yields the Poincaré inequality

$$-\langle\phi, Q_O\phi\rangle_{L^2(\mu_q)} = \beta^{-1}\|\nabla_q\phi\|^2_{L^2(\mu_q)} \geq m\,\|\phi - \textstyle\int\phi\,d\mu_q\|^2_{L^2(\mu_q)}. \tag{70}$$

Hence

$$\ker(Q_O) = \mathbb{C}\mathbf{1}, \qquad \lambda(Q_O) = m. \tag{71}$$

Since F is not a gradient field, the drift $\nabla_q U - \eta F(q)$ is not of gradient form for $\eta \neq 0$, so the effective generator $\mathcal{L}_O$ is genuinely nonequilibrium.

Non-self-adjointness on the slow space. In $L^2(\mu_q)$, integration by parts gives

$$(F\cdot\nabla_q)^\dagger = -F\cdot\nabla_q - (\nabla\cdot F) + \beta\,\nabla_q U\cdot F. \tag{72}$$

Hence $\mathcal{L}_O$ is generically non-self-adjoint as soon as $\eta \neq 0$. For the concrete force used in the numerics below,

$$F(q) = (-\tanh q_2, \tanh q_1), \tag{73}$$

one has $\nabla\cdot F = 0$ but $\nabla_q U\cdot F \not\equiv 0$ for any non-radial U, so $\mathcal{L}_O$ is explicitly non-self-adjoint for every $\eta \neq 0$.

Verification of Assumptions 4.1 and 4.2. The structural hypotheses hold with constants uniform in m and depending only on $(\beta, d, \|F\|_{L^\infty})$.

Proposition 6.2 (Verification for the convex-potential family)**.** *For the family* (67) *with* $U \in C^\infty(\mathbb{R}^d)$ *satisfying* $\nabla^2 U \geq m > 0$ *and with all derivatives of positive order growing at most polynomially, and bounded non-gradient force* F*, Assumptions 4.1, 4.2, and 4.3 hold in* $L^2(\mu)$*. Moreover, one may choose structural constants satisfying*

$$\lambda_R = 1, \qquad M_R = 1, \qquad \kappa_{\mathrm{ell}} = \sqrt{2}, \tag{74}$$

and the perturbation constants (a_1, b_1, a_2, b_2) *depend only on* $(\beta, d, \|F\|_{L^\infty})$ *and are independent of* m*. The extra polynomial-growth assumptions on the derivatives of* U *enter only in the qualitative verification of Assumption 4.3; they do not affect these quantitative constants.*

Corollary 6.3 (Rigorous kinetic scaling for the family)**.** *For the convex-potential Langevin family above, the hypotheses of Corollary 5.2 are satisfied uniformly as* $m \downarrow 0$*. Consequently,*

$$\gamma_{\mathrm{opt}} \asymp \sqrt{m}, \qquad \max_{\gamma>0} \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{m}, \qquad \eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{m}. \tag{75}$$

Thus this family gives a fully rigorous classical nonequilibrium verifier of the $\sqrt{\lambda(Q_O)}$ *acceleration mechanism within the present theorem.*

The proof of Proposition 6.2 is given in Appendix F. Hence Theorem 5.1 and Corollary 5.2 apply rigorously to this family.

Status of the ceiling. As noted above, the ceiling is conditional away from equilibrium in this $L^2(\mu)$ geometry, so Figure 1 should be read as numerical evidence for sharpness.

Numerical evidence on the truncated model. We verify the predicted scaling on the quadratic instance $U(q) = \frac{m}{2}|q|^2$ in $d = 2$, with $F(q) = (-\tanh q_2, \tanh q_1)$ and $\eta = c\sqrt{m}$, $c \in \{0, \frac{1}{2}, 1\}$. The generator is discretized in a tensor-product Hermite–Galerkin basis of dimension 256. These computations are illustrative and are not used in the proofs.

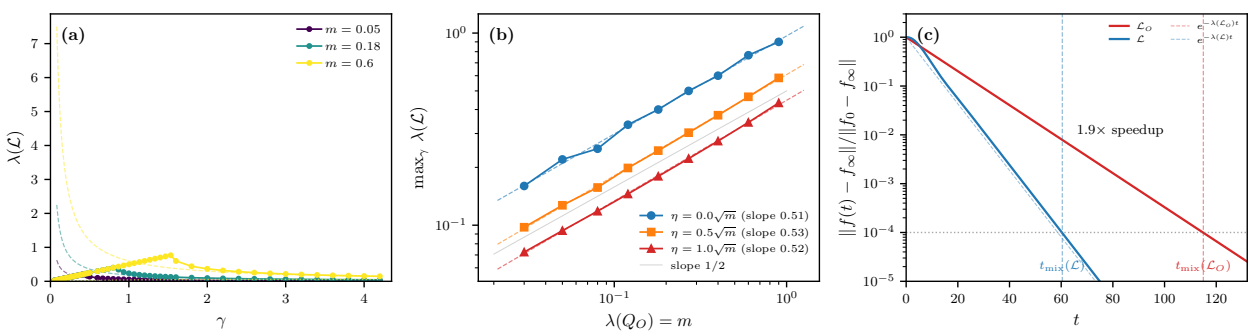

FIGURE 1. Numerical experiment on the truncated Hermite–Galerkin model for Example 6.2 ($\beta = 1$, $m \in [0.03, 0.90]$, 45-point γ-grid). **(a)** Spectral gap versus γ at $\eta = 0$; dashed: overdamped envelope m/γ; dotted: gap-adapted DMS lower bound. **(b)** Peak gap versus m on log–log scale for $c \in \{0, \frac{1}{2}, 1\}$; fitted slopes ≈ 0.51–0.53. **(c)** Semigroup decay at $m = 0.08$, $c = \frac{1}{2}$, $\gamma^* = 0.5$; vertical markers: ε-mixing times ($\varepsilon = 10^{-4}$).

Figure 1 shows the three-regime structure and the predicted $\sqrt{\lambda(Q_O)}$ scaling: the observed log–log slopes on the truncated matrices are 0.51, 0.53, and 0.52 for $c = 0$, $\frac{1}{2}$, 1. At $m = 0.08$, $c = \frac{1}{2}$, the truncated lifted gap $\lambda(\mathcal{L}) \approx 0.157$ gives about a 1.9× speedup over $\lambda(\mathcal{L}_O) \approx 0.080$, with mixing times $t_{\mathrm{mix}} \approx 60$ versus 115.

Algorithmic lift via generic square-root factorization. We apply the universal construction of Section 6.1 to the truncated slow Hermite space: $\mathcal{H}_{F,\mathrm{alg}} = \mathcal{H}_S$, $\mathcal{R}_{F,\mathrm{alg}} = -\mathrm{id}$, $T = (-Q_O)^{1/2}$, yielding a 32×32 algorithmic lift instead of the 256×256 physical lift. Since $\mathcal{V}_{\mathrm{pert}} P_S = 0$ and

$P_F \mathcal{V}_{\text{pert}} P_F = 0$, the consistency relation simplifies and Proposition 6.1 applies. Figure 2 and Table 1 show that the algorithmic and physical lifts track each other closely: fitted log–log slopes are 0.50–0.51 for both, and at $m = 0.08$, $c = \frac{1}{4}$ the mixing times are 47.5 (physical), 47.6 (algorithmic), and 115.2 (overdamped).

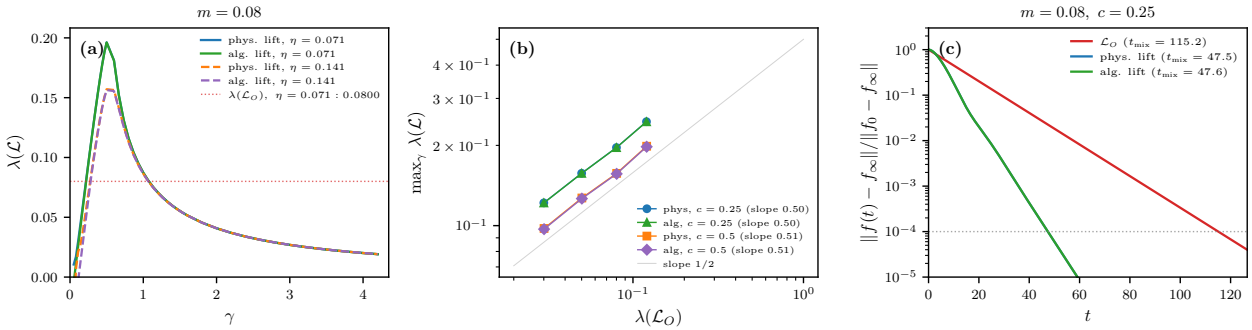

FIGURE 2. Comparison of the *physical* (256×256) and *algorithmic* (32×32, square-root factorization) lifted generators for the truncated model from Example 6.2. **(a)** Spectral gap versus γ at $m = 0.08$ for both lifts with $c = \frac{1}{4}$ and $c = \frac{1}{2}$ (that is, $\eta = c\sqrt{m}$); the overdamped reference is $\lambda(\mathcal{L}_O)$. **(b)** Log–log plot of peak spectral gap versus $\lambda(\mathcal{L}_O)$ on the matched grid $m \in \{0.03, 0.05, 0.08, 0.12\}$ for both lifts. **(c)** Convergence comparison at $m = 0.08$, $c = \frac{1}{4}$, $\varepsilon = 10^{-4}$: algorithmic lift, physical lift, and overdamped $\mathcal{L}_O$.

TABLE 1. Numerically computed peak spectral gaps $\max_\gamma \lambda(\mathcal{L})$ for the physical and algorithmic lifts on the matched grid. The final column reports the physical-lift speedup $\max_\gamma \lambda(\mathcal{L}_{\text{phys}})/\lambda(\mathcal{L}_O)$.

m	c	$\lambda(\mathcal{L}_O)$	phys. peak	alg. peak	phys. speedup
0.03	$\frac{1}{4}$	0.03001	0.12175	0.12163	4.06×
0.03	$\frac{1}{2}$	0.03010	0.09757	0.09701	3.24×
0.05	$\frac{1}{4}$	0.05001	0.15718	0.15703	3.14×
0.05	$\frac{1}{2}$	0.05016	0.12696	0.12629	2.53×
0.08	$\frac{1}{4}$	0.08002	0.19628	0.19613	2.45×
0.08	$\frac{1}{2}$	0.08024	0.15707	0.15640	1.96×
0.12	$\frac{1}{4}$	0.12002	0.24546	0.24519	2.05×
0.12	$\frac{1}{2}$	0.12032	0.19855	0.19765	1.65×

6.3. **Example 2: A Boundary-Driven Two-Qubit Model.** We now study a two-qubit boundary-driven model. Here "boundary-driven" refers to reservoir driving in a finite-dimensional open quantum system, not to kinetic boundary conditions in physical space. This is the finite-dimensional example: Theorem 5.1 is verified completely, the reduced slow dynamics is already nonequilibrium for $\eta \neq 0$, and all operators are bounded because $\dim \mathcal{H} = 16$. As in the Langevin example, the ceiling remains conditional away from equilibrium in the working Hilbert–Schmidt/KMS norm.

Setup. Both fast and slow subsystems are qubits: $\mathfrak{h}_F \cong \mathfrak{h}_S \cong \mathbb{C}^2$. We work in $\mathcal{H} = L^2(\sigma)$ with $\sigma = \frac{1}{2}\mathbf{1}_F \otimes \frac{1}{2}\mathbf{1}_S$. The slow subspace is $\mathcal{H}_S = \mathbf{1}_F \otimes \mathcal{B}(\mathfrak{h}_S)$, with projection $P_S(X) = \mathbf{1}_F \otimes \mathrm{tr}_F(\frac{1}{2}\mathbf{1}_F \cdot X)$.

Generator decomposition. The decomposition is $\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\text{pert}}$. Here $\mathcal{R} = \mathcal{R}_F \otimes \mathrm{id}_S$ with $\mathcal{R}_F X = \Gamma \sum_{j \in \{x,y,z\}} \mathcal{D}^\dagger[\sigma_j]X$, so $\lambda_R = 4\Gamma$. The transport part uses a *two-channel* coupling,

$$\mathcal{V}_{\text{skew}} X = -i[g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z), X], \tag{76}$$

and the perturbation is $\mathcal{V}_{\text{pert}} = \sigma_z \otimes \mathcal{G}_S$, $\mathcal{G}_S(X) = \mathcal{D}^\dagger[\sigma_-]X$. Because $\operatorname{tr}(\frac{1}{2}\mathbf{1}_F \cdot \sigma_z) = 0$, both terms map $\mathcal{H}_S$ into $\mathcal{H}_F$, so centering is satisfied.

Effective generator $\mathcal{L}_O$ (explicit formula). Let $X = \mathbf{1}_F \otimes X_S \in \mathcal{H}_S$. Because all operators are bounded in this 16×16 model, the three-step computation below is an exact evaluation of Definition 3.4, not a formal heuristic.

Step 1 (first excursion to fast space). The two-channel coupling gives

$$(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X = \sum_{a\in\{x,z\}} \sigma_a \otimes \Big(- ig\,[\sigma_a, X_S]\Big) + \sigma_z \otimes \eta\,\mathcal{G}_S(X_S). \tag{77}$$

Step 2 (Poisson equation on the fast ancilla). Since $\mathcal{R}_F(\sigma_a) = -\kappa\,\sigma_a$ with $\kappa := 4\Gamma$ for each $a \in \{x, y, z\}$, we have $\mathcal{R}^+(\sigma_a \otimes Y) = -(1/\kappa)\sigma_a \otimes Y$. Therefore

$$\mathcal{R}^+(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X = -\frac{1}{\kappa}\sum_{a\in\{x,z\}} \sigma_a \otimes \Big(- ig\,[\sigma_a, X_S]\Big) - \frac{1}{\kappa}\sigma_z \otimes \eta\,\mathcal{G}_S(X_S). \tag{78}$$

Step 3 (return map and projection). Using $\mathcal{V}_{\text{skew}}(\sigma_a \otimes Y) = -ig\sum_{b\in\{x,z\}} \sigma_a\sigma_b \otimes [\sigma_b, Y]$, the cross-channel products $\sigma_x\sigma_z = -i\sigma_y$ and $\sigma_z\sigma_x = i\sigma_y$ produce σ_y factors, which vanish under P_S (since $\operatorname{tr}(\sigma_y) = 0$). Only same-channel terms survive. Projecting gives

$$\begin{aligned}\mathcal{L}_O(\mathbf{1}_F \otimes X_S) = -\frac{1}{\kappa}\,\mathbf{1}_F \otimes \Big(& g^2 \sum_{a\in\{x,z\}} [\sigma_a, [\sigma_a, X_S]] \\ &+ i\eta g\,\big[\sigma_z, \mathcal{G}_S(X_S)\big] \\ &+ i\eta g\,\mathcal{G}_S([\sigma_z, X_S]) \\ &- \eta^2\,\mathcal{G}_S^2(X_S)\Big).\end{aligned} \tag{79}$$

The reference operator is the $\eta = 0$ part,

$$Q_O(\mathbf{1}_F \otimes X_S) = -\frac{g^2}{\kappa}\,\mathbf{1}_F \otimes \sum_{a\in\{x,z\}} [\sigma_a, [\sigma_a, X_S]]. \tag{80}$$

Non-self-adjointness on the slow space. Since the slow marginal of σ is proportional to the identity, the $L^2(\sigma)$ inner product on $\mathcal{H}_S$ reduces to the Hilbert–Schmidt inner product, and $(\mathcal{D}^\dagger[\sigma_-])^\dagger = \mathcal{D}[\sigma_-] \neq \mathcal{D}^\dagger[\sigma_-]$. Hence the $\mathcal{O}(\eta)$ correction in $\mathcal{L}_O - Q_O$ is already non-self-adjoint for every $\eta \neq 0$.

This can be seen explicitly on the Pauli basis. Using

$$\mathcal{G}_S(\mathbf{1}) = 0, \qquad \mathcal{G}_S(\sigma_x) = -\tfrac{1}{2}\sigma_x, \qquad \mathcal{G}_S(\sigma_y) = -\tfrac{1}{2}\sigma_y, \qquad \mathcal{G}_S(\sigma_z) = -(\mathbf{1} + \sigma_z), \tag{81}$$

the effective generator (79) acts as

$$\begin{aligned}\mathcal{L}_O(\mathbf{1}) &= 0,\\ \mathcal{L}_O(\sigma_x) &= -\frac{4g^2 - \eta^2/4}{\kappa}\,\sigma_x - \frac{2\eta g}{\kappa}\,\sigma_y,\\ \mathcal{L}_O(\sigma_y) &= \frac{2\eta g}{\kappa}\,\sigma_x - \frac{8g^2 - \eta^2/4}{\kappa}\,\sigma_y,\\ \mathcal{L}_O(\sigma_z) &= \frac{\eta^2}{\kappa}\,\mathbf{1} - \frac{4g^2 - \eta^2}{\kappa}\,\sigma_z.\end{aligned} \tag{82}$$

The σ_x–σ_y coupling and the $\sigma_z \to \mathbf{1}$ drift vanish when $\eta = 0$; for $\eta \neq 0$ they provide an explicit algebraic witness that the reduced dynamics itself is already away from detailed balance.

Verification of Assumptions 4.1 and 4.2. All operators are bounded on the finite-dimensional space $\mathcal{H}$, so the verification reduces to linear algebra.

Proposition 6.4 (Verification for the boundary-driven model)**.** *For the boundary-driven two-qubit family above with the two-channel coupling* $H_{\text{int}} = g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z)$*, Assumptions 4.1 and 4.2 hold. Moreover,*

$$\lambda_R = 4\Gamma, \qquad M_R = \|\mathcal{R}\|, \qquad \ker(Q_O) = \mathbb{C}\mathbf{1}, \qquad \lambda(Q_O) = g^2/\Gamma, \tag{83}$$

and the perturbation constants (a_1, b_1, a_2, b_2) *satisfy* $a_1 = a_2 = 0$*,* $b_j = \|\mathcal{K}_{\text{pert}}\|$ *or* $\|\mathcal{S}_{\text{pert}}\|$*.*

The proof is given in Appendix G. Hence Theorem 5.1 applies rigorously to the boundary-driven toy.

Status of the ceiling. As in Example 6.2, the ceiling is conditional away from equilibrium, so Figure 3 should be read as numerical evidence for sharpness rather than as a model-specific proof.

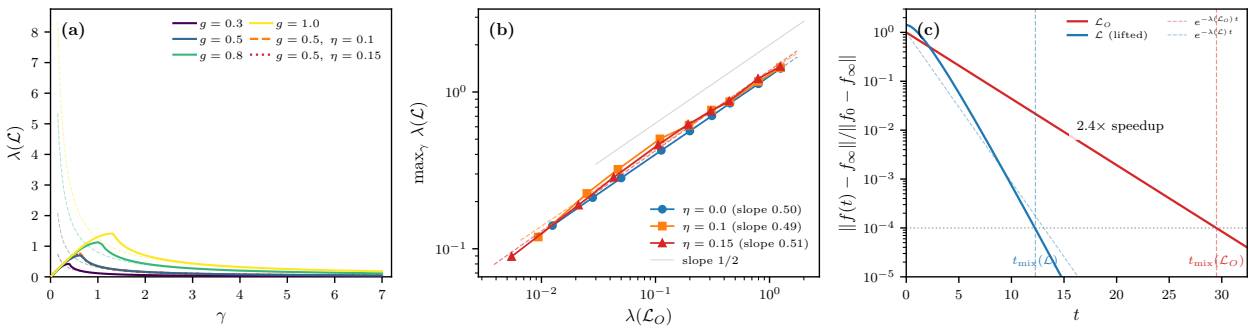

FIGURE 3. Numerical verification for Example 6.3. **(a)** Spectral gap $\lambda(\mathcal{L})$ vs. γ at $\eta = 0$ for several values of g; dashed curves show the overdamped envelope $\lambda(\mathcal{L}_O)/\gamma$. **(b)** Log–log plot of $\max_\gamma \lambda(\mathcal{L})$ versus $\lambda(\mathcal{L}_O)$. In this finite-dimensional setting the fitted slopes track the half-law closely, so the panel serves as a clean numerical benchmark for the mechanism. **(c)** Decay of the projected slow component for the lifted dynamics at γ^* and for $\mathcal{L}_O$ ($g = 0.5$, $\eta = 0$); dashed lines show the exponential envelopes. Vertical markers indicate the ε-level crossing times ($\varepsilon = 10^{-4}$). Parameters: $\Gamma = 0.8$, $\kappa = 3.2$, $g \in \{0.1, \ldots, 1.0\}$, 53-point γ-grid over $[0.02, 7.0]$.

TABLE 2. Kernel compatibility and spectral gap for Q_O with the two-channel coupling $H_{\text{int}} = g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z)$. For all $g > 0$: $\dim\ker(Q_O) = 1$ ($= \mathbb{C}\mathbf{1}$) and $\lambda(Q_O) = g^2/\Gamma$.

g	$\dim\ker(Q_O)$	$\lambda(Q_O)$
0.3	1	0.1125
0.5	1	0.3125
1.0	1	1.2500

Numerical evidence on the exact finite-dimensional model. We assemble the exact 16×16 generator and scan spectral gaps over $g \in \{0.1, \ldots, 1.0\}$, $\Gamma = 0.8$, $\gamma \in [0.02, 7.0]$, and $\eta \in \{0, 0.10, 0.15\}$. Figure 3 again shows the three-regime structure and the $\sqrt{\lambda(Q_O)}$ scaling: the fitted log–log slopes are 0.50, 0.49, and 0.51, and at $g = 0.5$, $\eta = 0$, the lifted gap yields about a 2.4× speedup.

Algorithmic lift via generic square-root factorization. We use the same universal square-root factorization as above, now in the boundary-driven toy model: $\mathcal{H}_{F,\text{alg}} = \mathcal{H}_S$, $\mathcal{R}_{F,\text{alg}} = -\lambda_R\,\text{id}$, and $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. Solving the consistency relation (60) numerically on the finite slow space gives

an 8×8 algorithmic lift (versus 16×16 for the physical model); the final residual is below 10^{-13}. Figure 4 and Table 3 show that the physical and algorithmic lifts track each other closely in both peak gaps and convergence curves, and that the stationary states agree to machine precision.

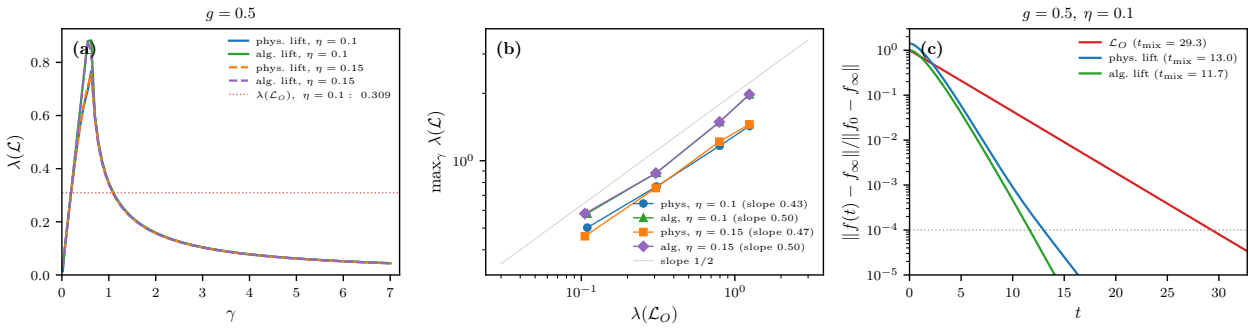

FIGURE 4. Comparison of the *physical* (16×16) and *algorithmic* (8×8, square-root factorization) lifted generators for the finite-dimensional model of Example 6.3. **(a)** Spectral gap $\lambda(\mathcal{L})$ vs. γ at $g = 0.5$ for both lifts ($\eta = 0.10$ and $\eta = 0.15$); the overdamped reference is $\lambda(\mathcal{L}_O)$. **(b)** Log–log plot of peak spectral gap vs. $\lambda(\mathcal{L}_O)$ for both lifts. **(c)** Convergence comparison ($g = 0.5$, $\eta = 0.10$, $\varepsilon = 10^{-4}$): algorithmic lift, physical lift, and overdamped $\mathcal{L}_O$.

TABLE 3. Numerically computed peak spectral gaps $\max_\gamma \lambda(\mathcal{L})$ for the physical and algorithmic lifts (53-point γ-grid). The "speedup" column is $\max_\gamma \lambda(\mathcal{L}_{\rm phys})/\lambda(\mathcal{L}_O)$.

g	η	$\lambda(\mathcal{L}_O)$	phys. peak	alg. peak	phys. speedup
0.3	0.10	0.1094	0.5024	0.5806	4.59×
0.3	0.15	0.1055	0.4600	0.5806	4.36×
0.5	0.10	0.3094	0.7673	0.8831	2.48×
0.5	0.15	0.3055	0.7552	0.8800	2.47×
0.8	0.10	0.7969	1.1663	1.4906	1.46×
0.8	0.15	0.7930	1.2163	1.4906	1.53×
1.0	0.10	1.2469	1.4291	1.9789	1.15×
1.0	0.15	1.2430	1.4534	1.9789	1.17×

For the *algorithmic* lifts in both examples, $P_F \mathcal{V}_{\rm pert} P_F = 0$ by construction, so Proposition 6.1 predicts zero projection bias. Numerically, $\|P_S h_\eta^{\rm alg} - g_\eta\|/\|g_\eta\| \lesssim 10^{-16}$ at all tested parameter values; the physical lifts give the same result on the adopted truncations.

7. Conclusion and open problems

The paper's central finding is that the spectral multiplier $\lambda/(\varpi + \lambda)$ in the DMS corrector is the bottleneck for kinetic acceleration, and choosing $\varpi \asymp \lambda(Q_O)$ removes it. The main certified output is the explicit lower bound of Theorem 5.1; the overdamped ceiling of Theorem 5.4 is the complementary conditional sharpness statement. Together they identify the kinetic $\sqrt{\lambda(Q_O)}$ scale for uniformly controlled families and isolate the main remaining gap: extending the ceiling beyond the contractive Hilbert-space setting.

(1) Sharp stability thresholds and non-convex potentials. The threshold η^* involves structural constants that may not be sharp; numerically, acceleration persists well beyond

η^*. The present convex-potential Langevin family (Section 6.2) establishes family-uniform structural constants for all U with $\nabla^2 U \geq m > 0$, but extending the result to non-convex potentials (where the Bochner inequality fails) remains open. For models where forcing enhances the effective gap (e.g. optimally rotated diffusions [18]), one expects $\lambda(\mathcal{L}_O) > \lambda(Q_O)$, and a refined analysis could yield

$$\max_{\gamma>0} \lambda(\mathcal{L}_\gamma) \asymp \sqrt{\lambda(\mathcal{L}_O)} \qquad \text{(conjectured, when } \lambda(\mathcal{L}_O) > \lambda(Q_O)\text{)}. \tag{84}$$

(2) Optimal factorization. The reverse adiabatic elimination recipe lifts any target Q_O, but the structural constants depend on the auxiliary space $(\mathcal{H}_F, \mathcal{R}_F, T)$. Characterizing the optimal factorization, and extending the framework to multilevel decompositions $\mathcal{H}_{S_1} \subset \mathcal{H}_{S_2} \subset \cdots$, remain open.

(3) Beyond the perturbative regime. Extending the analysis to large η may require space-time divergence methods [12, 13] or probabilistic contractivity arguments [27]. A related problem is to construct stationary-state-adapted Hilbert norms in which contraction holds and to track how λ_R, M_R, κ_{ell}, and the perturbation constants transform under that renorming. Alternatively, one could seek a ceiling in Harris-type or Wasserstein frameworks that avoids strict L^2 contraction.

(4) Kinetic boundary conditions. Our PDE example is posed on $\mathbb{R}^d$ with confining potential. Extending Assumption 4.1.3 to bounded domains with diffuse/specular reflection, grazing collisions, or Knudsen layers—where boundary layers may affect the constants M_R and κ_{ell}—remains open.

(5) Quantum many-body and infinite-dimensional NESS. The two-qubit toy model of Section 6.3 shows that the framework applies to finite-dimensional boundary-driven Lindbladians [15, 26] when suitable noncommutative Poincaré inequalities hold; compare [24]. Extending the framework to infinite-dimensional or many-body systems (e.g., boundary-driven XXZ chains) requires additional control of the KMS/modular domain and of the pseudoinverse in the corresponding noncommutative L^2 space, together with meaningful lower bounds on $\lambda(Q_O)$ and structural constants that remain informative with system size.

References

[1] Dominique Bakry and Michel Émery, *Diffusions hypercontractives*, Séminaire de probabilités, XIX, 1983/84, Lecture Notes in Math., vol. 1123, Springer, Berlin, 1985, pp. 177–206.

[2] Giovanni Brigati, Francis Lörler, and Lihan Wang, *Hypocoercivity meets lifts*, arXiv:2412.10890, 2025.

[3] Yu Cao, Jianfeng Lu, and Lihan Wang, *On explicit L^2-convergence rate estimate for underdamped Langevin dynamics*, Arch. Ration. Mech. Anal. **247** (2023), no. 5, 90.

[4] Eric A. Carlen, David A. Huse, and Joel L. Lebowitz, *Boundary-driven quantum systems near the Zeno limit: steady states and long-time behavior*, arXiv:2512.12825, 2025.

[5] Eric A. Carlen, David A. Huse, and Joel L. Lebowitz, *Stationary states of boundary driven quantum systems: some exact results*, Phys. Rev. A **111** (2025), no. 1, 012210.

[6] Sourav Chatterjee, *Spectral gap of nonreversible Markov chains*, arXiv:2310.10876, 2023.

[7] Fang Chen, László Lovász, and Igor Pak, *Lifting Markov chains to speed up mixing*, Proceedings of the Thirty-First Annual ACM Symposium on Theory of Computing (New York), STOC '99, ACM, 1999, pp. 275–281.

[8] Helge Dietert, *L^2-hypocoercivity for non-equilibrium kinetic equations*, arXiv:2310.13456, 2023.

[9] Jean Dolbeault, Clément Mouhot, and Christian Schmeiser, *Hypocoercivity for kinetic equations with linear relaxation terms*, arXiv:0810.3493, 2009.

[10] Jean Dolbeault, Clément Mouhot, and Christian Schmeiser, *Hypocoercivity for linear kinetic equations conserving mass*, Trans. Amer. Math. Soc. **367** (2015), no. 6, 3807–3828.

[11] Andreas Eberle, Arnaud Guillin, Leo Hahn, Francis Lörler, and Manon Michel, *Convergence of non-reversible Markov processes via lifting and flow Poincaré inequality*, arXiv:2503.04238, 2025.

[12] Andreas Eberle and Francis Lörler, *Non-reversible lifts of reversible diffusion processes and relaxation times*, arXiv:2402.05041, 2024.

[13] Andreas Eberle and Francis Lörler, *Space-time divergence lemmas and optimal non-reversible lifts of diffusions on Riemannian manifolds with boundary*, arXiv:2412.16710, 2024.

[14] Di Fang, Jianfeng Lu, and Yu Tong, *Mixing time of open quantum systems via hypocoercivity*, Phys. Rev. Lett. **134** (2025), 140405.
[15] Vittorio Gorini, Andrzej Kossakowski, and E. C. G. Sudarshan, *Completely positive dynamical semigroups of N-level systems*, J. Math. Phys. **17** (1976), no. 5, 821–825. MR 0413882
[16] Bernard Helffer and Francis Nier, *Hypoelliptic estimates and spectral theory for Fokker–Planck operators and Witten Laplacians*, Lecture Notes in Mathematics, vol. 1862, Springer, 2005.
[17] Frédéric Hérau, *Short and long time behavior of the Fokker–Planck equation in a confining potential and applications*, J. Funct. Anal. **244** (2007), no. 1, 95–118.
[18] Chii-Ruey Hwang, Sheng-Yao Hwang-Ma, and Shuenn-Jyi Sheu, *Accelerating diffusions*, Ann. Appl. Probab. **15** (2005), no. 2, 1433–1444.
[19] Alessandra Iacobucci, Stefano Olla, and Gabriel Stoltz, *Convergence rates for nonequilibrium Langevin dynamics*, arXiv:1702.03685, 2017.
[20] Marcelo Janovitch, Matteo Brunelli, and Patrick P. Potts, *Active quantum reservoir engineering: using a qubit to manipulate its environment*, arXiv:2505.16898, 2025.
[21] Tosio Kato, *Perturbation theory for linear operators*, reprint of the 1980 edition ed., Classics in Mathematics, Springer-Verlag, Berlin, 1995.
[22] E. M. Kessler, *Generalized Schrieffer-Wolff formalism for dissipative systems*, Phys. Rev. A **86** (2012), no. 1, 012126.
[23] Gabriel T. Landi, Dario Poletti, and Gernot Schaller, *Nonequilibrium boundary-driven quantum systems: Models, methods, and properties*, Rev. Mod. Phys. **94** (2022), 045006.
[24] Bowen Li and Jianfeng Lu, *Quantum space-time Poincaré inequality for Lindblad dynamics*, arXiv:2406.09115, 2024.
[25] Bowen Li and Jianfeng Lu, *Speeding up quantum Markov processes through lifting*, arXiv:2505.12187, 2025.
[26] G. Lindblad, *On the generators of quantum dynamical semigroups*, Comm. Math. Phys. **48** (1976), no. 2, 119–130. MR 0413878
[27] Pierre Monmarché, *L^2 geometric ergodicity for the kinetic Langevin process with non-equilibrium steady states*, arXiv:2501.18004, 2025.
[28] Pierre Monmarché and Mouad Ramil, *Overdamped limit at stationarity for non-equilibrium Langevin diffusions*, arXiv:2110.01238, 2022.
[29] Michael Reed and Barry Simon, *Methods of modern mathematical physics, vol. II: Fourier analysis, self-adjointness*, Academic Press, New York, 1975.
[30] Julien Roussel and Gabriel Stoltz, *Spectral methods for Langevin dynamics and associated error estimates*, ESAIM Math. Model. Numer. Anal. **52** (2018), no. 3, 1051–1083.
[31] Masaaki Tokieda and Angela Riva, *Time-convolutionless master equation applied to adiabatic elimination*, Phys. Rev. A **111** (2025), no. 5, 052206.
[32] Cédric Villani, *Hypocoercivity*, Mem. Amer. Math. Soc., vol. 202, American Mathematical Society, Providence, RI, 2009. MR 2562709

Appendix A. Analytic toolbox: pseudoinverse and Friedrichs construction

This appendix collects the standard functional-analytic constructions underlying the slow–fast framework. In finite dimensions every step is trivial; the statements below are needed only for the unbounded (PDE/diffusion) setting.

Lemma A.1 (Pseudoinverse via spectral calculus)**.** *Let $\mathcal{R}$ be a non-positive self-adjoint operator on $\mathcal{H}$ with $\ker(\mathcal{R}) = \mathcal{H}_S$ and spectral gap $\lambda_R > 0$ on $\mathcal{H}_F = \mathcal{H}_S^\perp$. Let $\{E_\lambda\}_{\lambda\geq 0}$ be the spectral family of $-\mathcal{R}|_{\mathcal{H}_F}$, so that $-\mathcal{R}|_{\mathcal{H}_F} = \int_{\lambda_R}^\infty \lambda\, dE_\lambda$. Then the pseudoinverse $\mathcal{R}^+ := -(-\mathcal{R}|_{\mathcal{H}_F})^{-1}P_F$ is a bounded self-adjoint operator on $\mathcal{H}$ satisfying:*

(i) $\mathcal{R}^+$ maps $\mathcal{H}_F$ into $\mathrm{dom}(\mathcal{R}) \cap \mathcal{H}_F$ and annihilates $\mathcal{H}_S$;
(ii) $\mathcal{R}\mathcal{R}^+ = P_F$ and $\mathcal{R}^+\mathcal{R} = P_F$ on $\mathrm{dom}(\mathcal{R})$;
(iii) $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$ and $-\mathcal{R}^+ \geq 0$ on $\mathcal{H}_F$;
(iv) the square root satisfies $\|(-\mathcal{R}^+)^{1/2}\| \leq \lambda_R^{-1/2}$.

Proof. Since the spectrum of $-\mathcal{R}|_{\mathcal{H}_F}$ is contained in $[\lambda_R, \infty)$, the inverse $(-\mathcal{R}|_{\mathcal{H}_F})^{-1} = \int_{\lambda_R}^\infty \lambda^{-1}\, dE_\lambda$ is a bounded non-negative self-adjoint operator on $\mathcal{H}_F$ with norm at most λ_R^{-1}. The remaining properties follow from the spectral theorem and the identities $E_\lambda \cdot \mathrm{dom}(\mathcal{R}) = \mathrm{dom}(\mathcal{R}) \cap E_\lambda\mathcal{H}$ for each spectral projection. □

Lemma A.2 (Friedrichs construction of the slow operator Q_O)**.** *Under the standing hypotheses on $\mathcal{R}$ and $\mathcal{V}_{\rm skew}$ (including the centered coupling condition $P_S\mathcal{V}_{\rm skew}P_S = 0$), and given a dense subspace $\mathcal{C} \subset \mathrm{dom}(\mathcal{V}_{\rm skew}) \cap \mathcal{H}_S$ that is invariant under $\mathcal{R}^+$ and $P_S\mathcal{V}_{\rm skew}$, define the quadratic form*

$$q_O(\psi) := \langle \mathcal{V}_{\rm skew}\psi,\, (-\mathcal{R}^+)\, \mathcal{V}_{\rm skew}\psi\rangle, \qquad \psi \in \mathcal{C}. \tag{85}$$

Then:

(i) q_O is a non-negative closed symmetric form on $\mathcal{H}_S$ (the closure being taken in the $\|\cdot\| + q_O(\cdot)^{1/2}$ graph topology).

(ii) The Friedrichs extension of q_O is a non-positive self-adjoint operator Q_O on $\mathcal{H}_S$ satisfying

$$\langle \psi, (-Q_O)\psi\rangle = q_O(\psi), \qquad \forall \psi \in \mathrm{dom}((-Q_O)^{1/2}). \tag{86}$$

(iii) On $\mathcal{C}$, this form identity reads $\langle \psi, (-Q_O)\psi\rangle = \langle \mathcal{V}_{\rm skew}\psi,\, (-\mathcal{R}^+)\, \mathcal{V}_{\rm skew}\psi\rangle$, recovering the operator expression (17)*.*

(iv) For every $\varpi > 0$, the resolvent $(\varpi - Q_O)^{-1}$ is a bounded self-adjoint operator on $\mathcal{H}_S$ with $\|(\varpi - Q_O)^{-1}\| \le \varpi^{-1}$. If $\lambda(Q_O) > 0$ (Assumption 4.2), then additionally $\|(\varpi - Q_O)^{-1}\| \le (\varpi + \lambda(Q_O))^{-1}$ on $\mathcal{H}_{S,0}$.

Proof. Symmetry follows from the skew-adjointness of $\mathcal{V}_{\rm skew}$:

$$q_O(\psi) = \langle \mathcal{V}_{\rm skew}\psi,\, (-\mathcal{R}^+)\, \mathcal{V}_{\rm skew}\psi\rangle \ge 0,$$

since $-\mathcal{R}^+ \ge 0$ on $\mathcal{H}_F$. The centering condition $P_S\mathcal{V}_{\rm skew}P_S = 0$ ensures that $\mathcal{V}_{\rm skew}\psi \in \mathcal{H}_F$ for $\psi \in \mathcal{H}_S$, so the form is well defined. Closedness is standard (the form norm $\|\psi\|^2 + q_O(\psi)$ controls the graph norm [21, Thm. VI.2.1]). The Friedrichs extension theorem [29, Thm. X.23] then provides the self-adjoint operator, and the resolvent bound is an immediate consequence of $\sigma(-Q_O) \subset [0, \infty)$ on $\mathcal{H}_S$. On $\mathcal{H}_{S,0}$, the spectrum of $-Q_O$ is contained in $[\lambda(Q_O), \infty)$, giving the sharper resolvent bound. □

Appendix B. Proofs for Preliminaries

This appendix collects the proofs from Section 3, carried out at Hilbert-space level with $\mathcal{H} = \mathbb{C}\mathbf{1} \oplus \mathcal{H}_0$.

B.1. **Proof of Lemma 3.2.** We first clarify the geometric relationship between the fixed point space $\mathcal{F}(\mathcal{L}) = \mathbb{C}\mathbf{1}$ and the kernel of the symmetric part $\mathcal{S}$.

Proposition B.1 (Kernel Inclusion)**.** *For any dissipative generator with symmetric part $\mathcal{S} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger) \le 0$, one has*

$$\ker(\mathcal{L}) \subseteq \ker(\mathcal{S}). \tag{87}$$

Proof. Let $X \in \ker(\mathcal{L})$. Then $\mathcal{L}(X) = 0$. Consequently,

$$\langle X, \mathcal{S}(X)\rangle = \Re\langle X, \mathcal{L}(X)\rangle = 0. \tag{88}$$

Since $\mathcal{S}$ is self-adjoint and non-positive, the operator $-\mathcal{S} \ge 0$ admits a square root. The identity above is equivalent to

$$\|(-\mathcal{S})^{1/2}X\|^2 = \langle X, (-\mathcal{S})X\rangle = 0.$$

Hence $(-\mathcal{S})^{1/2}X = 0$, and therefore $\mathcal{S}X = 0$. Thus $X \in \ker(\mathcal{S})$. □

Proof of Lemma 3.2. We prove the coercivity characterization and show that, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \ne \{0\}$, then no exponential decay estimate can hold with prefactor $C = 1$.

($\Leftarrow$): Assume $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$ and $-\mathcal{S}|_{\mathcal{H}_0}$ has a positive spectral gap $\lambda_S > 0$.

$$\langle X, \mathcal{S}X\rangle \le -\lambda_S\|X\|^2, \qquad \forall X \in \mathcal{H}_0. \tag{89}$$

For any $X_0 \in \mathrm{dom}(\mathcal{L}) \cap \mathcal{H}_0$, set $X_t = \mathcal{P}_t X_0$. Then $t \mapsto X_t$ is differentiable in $\mathcal{H}_0$, and the evolution of the norm satisfies:

$$\frac{1}{2}\frac{d}{dt}\|X_t\|^2 = \Re\langle X_t, \mathcal{L}X_t\rangle = \langle X_t, \mathcal{S}X_t\rangle \le -\lambda_S \|X_t\|^2. \tag{90}$$

Integrating this differential inequality yields $\|X_t\| \le e^{-\lambda_S t}\|X_0\|$ for all such X_0. By density of $\mathrm{dom}(\mathcal{L}) \cap \mathcal{H}_0$ in $\mathcal{H}_0$ and continuity of $\mathcal{P}_t$, the same estimate extends to every $X_0 \in \mathcal{H}_0$. This is an exponential decay with prefactor $C = 1$ and rate λ_S. Thus, the system is coercive, not strictly hypocoercive.

($\Rightarrow$): Assume the system is coercive ($C = 1$). By definition, there exists $\nu > 0$ such that $\|\mathcal{P}_t X\| \le e^{-\nu t}\|X\|$ for all $X \in \mathcal{H}_0$. For $X \in \mathrm{dom}(\mathcal{L}) \cap \mathcal{H}_0$, divide the squared estimate by $2t$ and let $t \downarrow 0$:

$$\Re\langle X, \mathcal{L}X\rangle \le -\nu\|X\|^2, \quad \forall X \in \mathrm{dom}(\mathcal{L}) \cap \mathcal{H}_0. \tag{91}$$

Substituting $\Re\langle X, \mathcal{L}X\rangle = \langle X, \mathcal{S}X\rangle$, we obtain:

$$\langle X, \mathcal{S}X\rangle \le -\nu\|X\|^2. \tag{92}$$

Now, suppose there exists a non-trivial vector $Y \in \ker(\mathcal{S}) \cap \mathcal{H}_0$. Then $\langle Y, \mathcal{S}Y\rangle = 0$. However, the inequality above requires $\langle Y, \mathcal{S}Y\rangle \le -\nu\|Y\|^2 < 0$, a contradiction. Therefore, $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$, which implies $\ker(\mathcal{S}) = \mathbb{C}\mathbf{1} = \ker(\mathcal{L})$.

Finally, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \ne \{0\}$ then the above argument shows that coercivity ($C = 1$ with some positive rate) is impossible. In that case, any exponential decay estimate (when it holds) must necessarily come with some prefactor $C > 1$. □

Appendix C. Proof of Lemma 4.6

This appendix establishes the structural bounds of Lemma 4.6.

Proof. In the unbounded setting, all identities below are first established for f in the common core $\mathcal{C}$ from Assumption 4.3; the resulting operator bounds then extend to the stated energy spaces by density and continuity. Recall the definition of the reference effective dynamics $Q_O = -P_S\mathcal{V}_{\mathrm{skew}}\mathcal{R}^+\mathcal{V}_{\mathrm{skew}}P_S$. By Assumption 4.1.1, the pseudoinverse is bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \le \lambda_R^{-1}$, hence

$$\langle g, -\mathcal{R}^+ g\rangle \le \frac{1}{\lambda_R}\|g\|^2, \quad \forall g \in \mathcal{H}_F. \tag{93}$$

The complementary (lower) control needed below is supplied by the flux conditioning assumption (22), which applies to fluxes of the form $g = g_\psi = P_F\mathcal{V}_{\mathrm{skew}}P_S\psi$.

C.1. **Proof of Properties 1 and 2 (Boundedness and Regularity). Solver boundedness and forward regularity.** Let $f \in \mathcal{H}$ and define $u := \mathcal{A}_\varpi f \in \mathcal{H}_S$. The defining equation is $(\varpi - Q_O)u = P_S\mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+)P_F f$. Taking the inner product with u and using the skew-adjointness $\mathcal{V}_{\mathrm{skew}}^\dagger = -\mathcal{V}_{\mathrm{skew}}$:

$$\varpi\|u\|^2 + \langle u, -Q_O u\rangle = \langle u, P_S\mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+)P_F f\rangle = -\langle \mathcal{V}_{\mathrm{skew}}u, (-\mathcal{R}^+)P_F f\rangle. \tag{94}$$

Let $v := \mathcal{V}_{\mathrm{skew}}u$. Since $P_S\mathcal{V}_{\mathrm{skew}}P_S = 0$ (Centered Coupling), we strictly have $v \in \mathcal{H}_F$. Using the definition of Q_O and the pseudoinverse bound (93):

$$\langle u, -Q_O u\rangle = \langle v, -\mathcal{R}^+ v\rangle = \|(-\mathcal{R}^+)^{1/2}v\|^2. \tag{95}$$

Substituting this into the energy identity and applying Cauchy–Schwarz yields:

$$\varpi\|u\|^2 + \|(-\mathcal{R}^+)^{1/2}v\|^2 \le \|(-\mathcal{R}^+)^{1/2}v\|\,\|(-\mathcal{R}^+)^{1/2}P_F f\|. \tag{96}$$

Writing $a := \|(-\mathcal{R}^+)^{1/2}v\|$ and $b := \|(-\mathcal{R}^+)^{1/2}P_F f\|$, we have $\varpi\|u\|^2 + a^2 \le ab$. Completing the square,

$$a^2 - ab = \left(a - \tfrac{1}{2}b\right)^2 - \tfrac{1}{4}b^2 \ge -\tfrac{1}{4}b^2, \tag{97}$$

so $\varpi\|u\|^2 \le \frac{1}{4}b^2$, hence

$$\|u\| \le \frac{1}{2}\,\varpi^{-1/2}\,\|(-\mathcal{R}^+)^{1/2}P_F f\| \le \frac{1}{2\sqrt{\lambda_R}}\,\varpi^{-1/2}\,\|f\|. \tag{98}$$

The first inequality is the spectral bound $\varpi^{-1/2}\cdot\frac{1}{2}$, which can equivalently be derived from the spectral calculus: writing $Q_O = -\int_0^\infty \lambda\, dE_\lambda$ on $\mathcal{H}_{S,0}$, one has $\|(\varpi - Q_O)^{-1}(-Q_O)^{1/2}\| = \sup_{\lambda\ge 0}\frac{\sqrt{\lambda}}{\varpi+\lambda}$, and the calculus identity $\sup_{\lambda\ge 0}\sqrt{\lambda}/(\varpi+\lambda) = 1/(2\sqrt{\varpi})$ (attained at $\lambda = \varpi$) gives the stated bound. The second inequality uses $\|(-\mathcal{R}^+)^{1/2}P_F f\| \le \lambda_R^{-1/2}\|f\|$. Moreover, $a \le b$. By construction of the resolvent, $u \in \mathrm{dom}(Q_O)$; in the unbounded setting this is exactly the mapping property required in Assumption 4.3. Then the vector $v = \mathcal{V}_{\mathrm{skew}}u$ is a slow-induced flux of the form g_ψ with $\psi \in \mathrm{dom}(Q_O)$; we may therefore apply the flux conditioning bound (22) to v to obtain

$$\|v\|^2 \le M_R\,\langle v, -\mathcal{R}^+ v\rangle = M_R a^2 \le M_R b^2 \le \frac{M_R}{\lambda_R}\,\|f\|^2, \tag{99}$$

so $\|v\| \le \sqrt{M_R/\lambda_R}\,\|f\|$. This proves $\|\mathcal{A}_\varpi\| \le k_{\mathcal{A}}\varpi^{-1/2}$ and $\|\mathcal{V}_{\mathrm{skew}}\mathcal{A}_\varpi\| \le k_{\mathrm{reg}}$, with

$$k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}, \qquad k_{\mathrm{reg}} := \sqrt{M_R/\lambda_R}. \tag{100}$$

Dual regularity. We consider the operator $\mathcal{T} := \mathcal{A}_\varpi\mathcal{V}_{\mathrm{skew}}P_F$. Its norm equals that of its adjoint $\mathcal{T}^\dagger$. We compute the adjoint of the solver:

$$\mathcal{A}_\varpi^\dagger = \left((\varpi - Q_O)^{-1}P_S\mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+)P_F\right)^\dagger = -P_F(-\mathcal{R}^+)\mathcal{V}_{\mathrm{skew}}P_S(\varpi - Q_O)^{-1}. \tag{101}$$

(Using $Q_O^\dagger = Q_O$ and $\mathcal{V}_{\mathrm{skew}}^\dagger = -\mathcal{V}_{\mathrm{skew}}$). Therefore:

$$\mathcal{T}^\dagger = P_F\mathcal{V}_{\mathrm{skew}}^\dagger\mathcal{A}_\varpi^\dagger = P_F\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)\mathcal{V}_{\mathrm{skew}}P_S(\varpi - Q_O)^{-1}. \tag{102}$$

Let $g \in \mathcal{H}_S$ and $\psi := (\varpi - Q_O)^{-1}g$. Define the macroscopic flux

$$g_\psi := P_F\mathcal{V}_{\mathrm{skew}}P_S\psi \in \mathcal{H}_F. \tag{103}$$

Then $\mathcal{T}^\dagger g = P_F\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)g_\psi$. Invoking the **Flux Regularity** condition from Assumption 4.1.3:

$$\|P_F\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)g_\psi\| \le \kappa_{\mathrm{ell}}\|(-Q_O)\psi\|. \tag{104}$$

Substituting ψ back:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\mathrm{ell}}\|(-Q_O)(\varpi - Q_O)^{-1}g\|. \tag{105}$$

Since Q_O is self-adjoint and non-positive, the operator $(-Q_O)(\varpi - Q_O)^{-1}$ is a spectral contraction: by functional calculus, $\|(-Q_O)(\varpi - Q_O)^{-1}\| = \sup_{\lambda\ge 0}\lambda/(\varpi+\lambda) \le 1$ (the supremum is ≤ 1 for every $\varpi > 0$, with equality in the limit $\lambda \to \infty$). Thus:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\mathrm{ell}}\|g\|. \tag{106}$$

This proves $\|\mathcal{A}_\varpi\mathcal{V}_{\mathrm{skew}}P_F\| \le \kappa_{\mathrm{ell}}$.

C.2. **Proof of Property 3 (Perturbation Control).** We use the structural decomposition $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ from Assumption 4.1.4. We also recall the centered mapping property (36), which will be used later to sharpen the perturbation bookkeeping in the dissipation matrix. Recall the bound for the skew part (Assumption 4.1.4(a)):

$$\|\mathcal{K}_{\text{pert}} f\| \le a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|, \qquad f \in \operatorname{dom}((-\mathcal{R})^{1/2}). \tag{107}$$

For the symmetric part, we use the energy bound (Assumption 4.1.4(b)):

$$\|\mathcal{S}_{\text{pert}} f\| \le a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|, \qquad f \in \operatorname{dom}((-\mathcal{R})^{1/2}). \tag{108}$$

Reverse estimate ($\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi$). Let $f \in \mathcal{H}$ and $u := \mathcal{A}_\varpi f$. The adjoint is $\mathcal{V}_{\text{pert}}^\dagger = -\mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$. Applying the triangle inequality and the bounds above:

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le \|\mathcal{K}_{\text{pert}} u\| + \|\mathcal{S}_{\text{pert}} u\| \tag{109}$$

$$\le b_1 \|u\| + b_2 \|u\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} u\|. \tag{110}$$

Since $u \in \mathcal{H}_S = \ker(\mathcal{R})$ (by definition of $\mathcal{A}_\varpi$), one has $(-\mathcal{R})^{1/2} u = 0$, hence

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le (b_1 + b_2) \|u\|. \tag{111}$$

Substituting the Solver Bound ($\|u\| \le k_{\mathcal{A}} \varpi^{-1/2} \|f\|$):

$$\|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi f\| \le (b_1 + b_2) k_{\mathcal{A}} \varpi^{-1/2} \|f\|. \tag{112}$$

This establishes (33a) with $k_P(\varpi) := (b_1 + b_2) k_{\mathcal{A}} \varpi^{-1/2}$.

Forward estimate ($\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}$). We aim to bound the operator norm of $\mathcal{W}_{\text{fwd}} := \mathcal{A}_\varpi \mathcal{V}_{\text{pert}}$. By adjoints on Hilbert spaces, $\|\mathcal{W}_{\text{fwd}}\| = \|\mathcal{W}_{\text{fwd}}^\dagger\| = \|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi^\dagger\|$. Let $g \in \mathcal{H}_S$ be arbitrary with $\|g\| = 1$ and set $w := \mathcal{A}_\varpi^\dagger g$. Using the bounds on $\mathcal{K}_{\text{pert}}$ and the energy bound on $\mathcal{S}_{\text{pert}}$,

$$\|\mathcal{V}_{\text{pert}}^\dagger w\| \le \|\mathcal{K}_{\text{pert}} w\| + \|\mathcal{S}_{\text{pert}} w\|$$

$$\le (b_1 + b_2) \|w\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} w\|. \tag{113}$$

We compute the norms on the right-hand side.

(1) **Solver term ($\|w\|$):**

$$\|w\| = \|\mathcal{A}_\varpi^\dagger g\| \le \|\mathcal{A}_\varpi\| \|g\| \le k_{\mathcal{A}} \varpi^{-1/2}. \tag{114}$$

(2) **Energy term ($\|(-\mathcal{R})^{1/2} w\|$):** Using $w = \mathcal{A}_\varpi^\dagger g$ and the functional calculus for the non-positive self-adjoint operator Q_O,

$$\|(-\mathcal{R})^{1/2} w\| = \|(-\mathcal{R})^{1/2} \mathcal{A}_\varpi^\dagger g\| = \|(-\mathcal{R})^{-1/2} P_F \mathcal{V}_{\text{skew}} P_S (\varpi - Q_O)^{-1} g\|. \tag{115}$$

Setting $\psi := (\varpi - Q_O)^{-1} g \in \mathcal{H}_S$, we note the identity (for such slow-induced fluxes)

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi, \tag{116}$$

$$\|(-\mathcal{R})^{-1/2} g_\psi\|^2 = \langle g_\psi, -\mathcal{R}^+ g_\psi \rangle = -\langle \psi, Q_O \psi \rangle = \|(-Q_O)^{1/2} \psi\|^2, \tag{117}$$

hence

$$\|(-\mathcal{R})^{1/2} w\| = \|(-Q_O)^{1/2} (\varpi - Q_O)^{-1} g\| \le \frac{1}{2} \varpi^{-1/2} \|g\|. \tag{118}$$

(The last step follows from the scalar bound $\sup_{x \ge 0} \frac{\sqrt{x}}{\varpi + x} = \frac{1}{2\sqrt{\varpi}}$.)

Substituting these estimates into (113) yields

$$\|\mathcal{V}_{\mathrm{pert}}^{\dagger}\mathcal{A}_{\varpi}^{\dagger}g\| \le \Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\,\varpi^{-1/2}\,\|g\|. \tag{119}$$

Since g is arbitrary, this establishes (33b) with $\tilde{k}_P(\varpi) := \big((b_1+b_2)k_{\mathcal{A}} + \frac{a_1+a_2}{2}\big)\varpi^{-1/2}$.

□

Appendix D. Proof of Theorem 5.1

This appendix derives the dissipation matrix governing the hypocoercive convergence. The computation is based on the modified Lyapunov functional with resolvent shift $\varpi > 0$ and mixing weight $\alpha > 0$:

$$\mathcal{E}_{\alpha,\varpi}(f) := \frac{1}{2}\|f\|^2 + \alpha\Re\langle \mathcal{A}_{\varpi}f, f\rangle, \tag{120}$$

where $\mathcal{A}_{\varpi} = (\varpi - Q_O)^{-1}P_S\mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+)P_F$ is the DMS corrector associated with the reference effective dynamics Q_O.

D.1. **Preliminaries.** This appendix uses the structural estimates of Lemma 4.6 (solver bound, regularity, and perturbation control) and the notation of equations (40)–(44). We set $\varpi := \varpi_* = \lambda(Q_O)$ throughout and use the effective perturbation coefficient

$$b_{2,\mathrm{eff}} := b_2 + \frac{a_2}{2}(1+\lambda_R), \tag{121}$$

as defined in (40). We also record the energy-admissibility threshold

$$\eta_{\mathrm{sym}}(\gamma) := \begin{cases} +\infty, & a_2 = 0,\\ 2\gamma/a_2, & a_2 > 0,\end{cases} \tag{122}$$

which is exactly the range for which the coefficient of $\|(-\mathcal{R})^{1/2}f\|^2$ in the basic energy estimate below remains nonnegative.

D.2. **Derivation of Differential Estimates. Standing regularity for the Lyapunov computation.** In the bounded-operator setting, for any $f_0 \in \mathcal{H}$ the solution $f_t = e^{t\mathcal{L}}f_0$ is continuously differentiable and satisfies $\partial_t f_t = \mathcal{L}f_t$. In unbounded settings, the algebraic identities below are first derived for classical trajectories $t \mapsto f_t$ taking values in the common core $\mathcal{C}$ and satisfying $\partial_t f_t = \mathcal{L}f_t$ there; this is the only place where model-dependent regularity enters. Once the differential inequality has been established on that regular class, the resulting semigroup estimate extends to arbitrary $f_0 \in \mathcal{H}_0$ by density together with the norm equivalence from Proposition 4.8 and continuity of the semigroup.

Lemma D.1 (Energy identity for the fast dissipator)**.** *Assume $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$. Then for all $f \in \mathrm{dom}(\mathcal{R})$,*

$$\Re\langle \mathcal{R}f, f\rangle = \langle \mathcal{R}f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2. \tag{123}$$

In particular, if $f = f_S + f_F$ with $f_S \in \ker(\mathcal{R})$ and $f_F \in \mathcal{H}_F$, then

$$\Re\langle \mathcal{R}f, f\rangle = \Re\langle \mathcal{R}f_F, f_F\rangle. \tag{124}$$

Proof. Since $\mathcal{R}$ is self-adjoint and non-positive, $-\mathcal{R} \ge 0$ admits a functional calculus square root $(-\mathcal{R})^{1/2}$ and $\langle \mathcal{R}f, f\rangle = -\langle(-\mathcal{R})f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2$. Moreover $\mathcal{R}f_S = 0$ and $\mathcal{R}f = \mathcal{R}f_F$. □

Let f_t denote such a classical solution. By differentiating (38) along the flow and using that $\mathcal{A}_\varpi$ is time-independent, we obtain

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\varpi}(f_t) = \Re\langle \partial_t f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi \partial_t f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f_t, \partial_t f_t\rangle \tag{125}$$

$$= \Re\langle \mathcal{L} f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L} f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f_t, \mathcal{L} f_t\rangle. \tag{126}$$

We now split the right-hand side into the "plain" dissipation coming from the generator itself and the hypocoercive correction generated by the cross term:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\varpi}(f_t) = \mathcal{D}_I(f_t) + \mathcal{D}_{\mathrm{mod}}(f_t), \tag{127}$$

where we define

$$\mathcal{D}_I(f) := \Re\langle \mathcal{L} f, f\rangle, \tag{128}$$

$$\mathcal{D}_{\mathrm{mod}}(f) := \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{L} f\rangle. \tag{129}$$

Expanding $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ in $\mathcal{D}_I$ then yields the decomposition displayed below.

Natural dissipation ($\mathcal{D}_I$).

$$\mathcal{D}_I = \gamma\Re\langle \mathcal{R} f, f\rangle + \Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle + \eta\Re\langle \mathcal{V}_{\mathrm{pert}} f, f\rangle. \tag{130}$$

Using Assumption 4.1.1 ($\mathcal{R}$ coercivity) and Assumption 4.1.4 (Perturbation decomposition $\mathcal{V}_{\mathrm{pert}} = \mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}$): the fast dissipator contributes $\gamma\Re\langle \mathcal{R} f, f\rangle \le -\gamma\lambda_R \|f_F\|^2$, while $\Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle = 0$ and $\Re\langle \mathcal{K}_{\mathrm{pert}} f, f\rangle = 0$ by skew-adjointness. For the symmetric part, using (27) and Cauchy–Schwarz,

$$\eta\Re\langle \mathcal{S}_{\mathrm{pert}} f, f\rangle \le \eta\|\mathcal{S}_{\mathrm{pert}} f\|\,\|f\| \le \eta\Big(b_2\|f\|^2 + a_2\,\|(-\mathcal{R})^{1/2} f\|\,\|f\|\Big) \tag{131}$$

$$\le \eta\Big(\big(b_2 + \tfrac{a_2}{2}\big)\|f\|^2 + \tfrac{a_2}{2}\,\|(-\mathcal{R})^{1/2} f\|^2\Big). \tag{132}$$

Combining with the dissipator contribution, note that by Lemma D.1, $\Re\langle \mathcal{R} f, f\rangle = \Re\langle \mathcal{R} f_F, f_F\rangle = -\|(-\mathcal{R})^{1/2} f\|^2$. If $\eta \le \eta_{\mathrm{sym}}(\gamma)$, i.e. if $\gamma - \eta a_2/2 \ge 0$, then we may use the lower bound $\|(-\mathcal{R})^{1/2} f\|^2 \ge \lambda_R \|f_F\|^2$ without changing the inequality direction. Hence

$$\gamma\Re\langle \mathcal{R} f, f\rangle + \eta\Re\langle \mathcal{S}_{\mathrm{pert}} f, f\rangle \le -\big(\gamma - \tfrac{\eta a_2}{2}\big)\,\|(-\mathcal{R})^{1/2} f\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f\|^2 \tag{133}$$

$$\le -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2, \tag{134}$$

where we used $\|(-\mathcal{R})^{1/2} f\|^2 \ge \lambda_R \|f_F\|^2$, the decomposition $\|f\|^2 = \|f_S\|^2 + \|f_F\|^2$, and the fact that $\frac{a_2\lambda_R}{2} + (b_2 + \frac{a_2}{2}) = b_{2,\mathrm{eff}}$ by definition. The fast mode carries the full effective coefficient $b_{2,\mathrm{eff}}$, while the slow mode retains only the smaller $b_2 + \frac{a_2}{2}$ (since $f_S \in \ker(\mathcal{R})$ does not benefit from the λ_R-amplification).

Thus:

$$\mathcal{D}_I \le -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2. \tag{135}$$

Hypocoercive correction ($\mathcal{D}_{\mathrm{mod}}$). Recall

$$\mathcal{D}_{\mathrm{mod}}(f) = \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{L} f\rangle. \tag{136}$$

Using $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ and linearity, we regroup $\mathcal{D}_{\mathrm{mod}}$ according to which part of the generator appears:

$$\mathcal{D}_{\mathrm{mod}}(f) = T_{\mathcal{R}}(f) + T_{\mathcal{V}}(f) + T_{\mathcal{P}}(f), \tag{137}$$

where

$$T_{\mathcal{R}}(f) := \alpha\gamma\,\Re\langle \mathcal{A}_\varpi \mathcal{R} f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\varpi f, \mathcal{R} f\rangle, \tag{138}$$

$$T_{\mathcal{V}}(f) := \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\mathrm{skew}} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\mathrm{skew}} f\rangle, \tag{139}$$

$$T_{\mathcal{P}}(f) := \alpha\eta\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\mathrm{pert}} f, f\rangle + \alpha\eta\,\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\mathrm{pert}} f\rangle. \tag{140}$$

Interaction with the fast dissipator. We start from the regrouped definition

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\varpi \mathcal{R} f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\varpi f, \mathcal{R} f\rangle. \tag{141}$$

Since $\mathcal{R} f = \mathcal{R} f_F$ and $\mathcal{A}_\varpi f \in \mathcal{H}_S = \ker(\mathcal{R})$, the second term vanishes by self-adjointness of $\mathcal{R}$:

$$\Re\langle \mathcal{A}_\varpi f, \mathcal{R} f\rangle = \Re\langle \mathcal{R}\mathcal{A}_\varpi f, f\rangle = 0. \tag{142}$$

Moreover $\mathcal{A}_\varpi \mathcal{R} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$, hence

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\varpi \mathcal{R} f_F, f_S\rangle. \tag{143}$$

Using $\mathcal{A}_\varpi = (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+) P_F$ and the pseudoinverse identity $\mathcal{R}^+\mathcal{R} = P_F$ on $\mathrm{dom}(\mathcal{R})$, we obtain

$$\mathcal{A}_\varpi \mathcal{R} f_F = (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+)\mathcal{R} f_F = -(\varpi - Q_O)^{-1} P_S \mathcal{V}_{\mathrm{skew}} f_F. \tag{144}$$

Therefore, by skew-adjointness of $\mathcal{V}_{\mathrm{skew}}$,

$$|T_{\mathcal{R}}| = \alpha\gamma\left|\Re\langle (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\mathrm{skew}} f_F, f_S\rangle\right| = \alpha\gamma\left|\Re\langle f_F, \mathcal{V}_{\mathrm{skew}}(\varpi - Q_O)^{-1} f_S\rangle\right| \tag{145}$$

$$\leq \alpha\gamma\,\|f_F\|\,\|\mathcal{V}_{\mathrm{skew}}(\varpi - Q_O)^{-1} f_S\|. \tag{146}$$

To bound $\|\mathcal{V}_{\mathrm{skew}}(\varpi - Q_O)^{-1} f_S\|$, set $\psi := (\varpi - Q_O)^{-1} f_S \in \mathrm{dom}(Q_O) \subset \mathcal{H}_S$ and denote its induced microscopic flux

$$g_\psi := P_F \mathcal{V}_{\mathrm{skew}} P_S \psi \in \mathcal{H}_F. \tag{147}$$

Since $\psi \in \mathcal{H}_S$, the centered-coupling assumption gives $P_S \mathcal{V}_{\mathrm{skew}} \psi = 0$, hence $g_\psi = \mathcal{V}_{\mathrm{skew}}\psi = \mathcal{V}_{\mathrm{skew}}(\varpi - Q_O)^{-1} f_S$. Applying the flux conditioning bound (22) yields

$$\|\mathcal{V}_{\mathrm{skew}}(\varpi - Q_O)^{-1} f_S\|^2 = \|g_\psi\|^2 \leq M_R\,\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle. \tag{148}$$

The "extra Q_O" comes from rewriting the right-hand side using the definition $Q_O := -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S$: indeed, for $\psi \in \mathcal{H}_S$,

$$\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle = \langle \mathcal{V}_{\mathrm{skew}}\psi, (-\mathcal{R}^+)\mathcal{V}_{\mathrm{skew}}\psi\rangle = \langle \psi, (-Q_O)\psi\rangle = \|(-Q_O)^{1/2}\psi\|^2. \tag{149}$$

Therefore

$$\|\mathcal{V}_{\mathrm{skew}}(\varpi - Q_O)^{-1} f_S\| \leq \sqrt{M_R}\,\|(-Q_O)^{1/2}(\varpi - Q_O)^{-1} f_S\|. \tag{150}$$

Finally, by functional calculus for the self-adjoint operator $-Q_O \geq 0$,

$$\|(-Q_O)^{1/2}(\varpi - Q_O)^{-1}\| = \sup_{x\geq 0} \frac{\sqrt{x}}{\varpi + x} = \frac{1}{2\sqrt{\varpi}}, \tag{151}$$

which implies

$$\|\mathcal{V}_{\mathrm{skew}}(\varpi - Q_O)^{-1} f_S\| \leq \frac{\sqrt{M_R}}{2}\,\varpi^{-1/2}\,\|f_S\|. \tag{152}$$

Hence

$$|T_{\mathcal{R}}| \leq \alpha\gamma\,\frac{\sqrt{M_R}}{2}\,\varpi^{-1/2}\,\|f_F\|\,\|f_S\|. \tag{153}$$

Interaction with transport. Starting from the regrouped definition

$$T_{\mathcal{V}}(f) = \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\rm skew} f\rangle, \tag{154}$$

we decompose $f = f_S + f_F$ and use that $\mathcal{A}_\varpi$ maps into the slow space and annihilates the slow component: $\mathcal{A}_\varpi = \mathcal{A}_\varpi P_F$ (since P_F is on the right in (30)). Hence $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$. Expanding the *first* bracket gives

$$\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle = \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \tag{155}$$

$$+ \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_F\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_F\rangle. \tag{156}$$

The last term vanishes because $\mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$. Moreover, since $\mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S \in \mathcal{H}_S$, we have $\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_F\rangle = 0$. Therefore

$$\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle = \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle. \tag{157}$$

In the *second* bracket we use $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$ and expand $\mathcal{V}_{\rm skew} f$:

$$\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\rm skew} f\rangle = \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle + \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle. \tag{158}$$

Putting the pieces together, we obtain the explicit decomposition

$$T_{\mathcal{V}} = \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \tag{159}$$

$$+ \alpha\,\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle. \tag{160}$$

Main Coercivity. Since $f_S \in \mathcal{H}_S$ and $P_S \mathcal{V}_{\rm skew} P_S = 0$, we have $P_F \mathcal{V}_{\rm skew} f_S = \mathcal{V}_{\rm skew} f_S$. By the definition of $\mathcal{A}_\varpi$ and Q_O this yields

$$\alpha\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle = \alpha\langle (\varpi - Q_O)^{-1} Q_O f_S, f_S\rangle = -\alpha\,\langle (\varpi - Q_O)^{-1}(-Q_O) f_S, f_S\rangle \tag{161}$$

$$\le -\alpha\,\frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)}\,\|f_S\|^2. \tag{162}$$

Here we used that $f \in \mathcal{H}_0$ implies $f_S = P_S f \in \mathcal{H}_S \cap \mathcal{H}_0 = \mathcal{H}_{S,0}$, so Assumption 4.2 applies to f_S.
Cross Terms. Since $\mathcal{A}_\varpi f_F \in \mathcal{H}_S$ (by the range of the corrector) and $\mathcal{V}_{\rm skew} f_S \in \mathcal{H}_F$ (by centering, $P_S \mathcal{V}_{\rm skew} P_S = 0$), the third inner product vanishes: $\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle = 0$. The only surviving cross term is bounded by the dual regularity estimate:

$$\alpha\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \le \alpha\|\mathcal{A}_\varpi \mathcal{V}_{\rm skew} P_F\|\|f_F\|\|f_S\| \le \alpha\kappa_{\rm ell}\|f_F\|\|f_S\|. \tag{163}$$

Fast–fast remainder. Using $\mathcal{V}_{\rm skew}^\dagger = -\mathcal{V}_{\rm skew}$ and Cauchy–Schwarz,

$$\big|\alpha\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle\big| = \alpha\big|\Re\langle -\mathcal{V}_{\rm skew}\mathcal{A}_\varpi f_F, f_F\rangle\big| \le \alpha\,\|\mathcal{V}_{\rm skew}\mathcal{A}_\varpi\|\,\|f_F\|^2 \le \alpha\,k_{\rm reg}\,\|f_F\|^2. \tag{164}$$

Interaction with perturbation.

$$T_{\mathcal{P}} = \alpha\eta\Re\langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f\rangle + \alpha\eta\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f\rangle. \tag{165}$$

We estimate each term using Lemma 4.6 and the centered mapping property (36).
Reverse term. Since $\mathcal{A}_\varpi f \in \mathcal{H}_S$ and $\mathcal{V}_{\rm pert}^\dagger \mathcal{H}_S \subset \mathcal{H}_F$, we have $\langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f\rangle = \langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f_F\rangle$. Therefore,

$$\big|\alpha\eta\Re\langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f\rangle\big| \le \alpha\eta\,\|\mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f\|\,\|f_F\| \le \alpha\eta\,k_P(\varpi)\,\|f\|\,\|f_F\| \tag{166}$$

$$\le \alpha\eta\,k_P(\varpi)\,\big(\|f_F\|^2 + \|f_F\|\,\|f_S\|\big). \tag{167}$$

Forward term. Since $\mathcal{A}_\varpi \mathcal{H} \subset \mathcal{H}_S$, we have $\langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f\rangle = \langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f_S\rangle$. Therefore,

$$\big|\alpha\eta\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f\rangle\big| \le \alpha\eta\,\|\mathcal{A}_\varpi \mathcal{V}_{\rm pert} f\|\,\|f_S\| \le \alpha\eta\,\tilde{k}_P(\varpi)\,\|f\|\,\|f_S\| \tag{168}$$

$$\le \alpha\eta\,\tilde{k}_P(\varpi)\,\big(\|f_S\|^2 + \|f_F\|\,\|f_S\|\big). \tag{169}$$

D.3. **Dissipation Matrix Construction.** Summing all estimates and using Cauchy–Schwarz bounds of the form $|\langle Xf_F, f_S\rangle| \le \|X\|\,\|f_F\|\,\|f_S\|$, we obtain a closed differential inequality of the form

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\varpi}(f_t) \le -\mathbf{v}(t)^T\mathbf{M}\,\mathbf{v}(t), \qquad \mathbf{v}(t) := \begin{pmatrix}\|P_F f_t\| \\ \|P_S f_t\|\end{pmatrix}. \tag{170}$$

Here $\mathbf{M}$ is a real symmetric 2×2 matrix whose entries depend only on the structural constants and the chosen parameters (α,ϖ). The entries of $\mathbf{M}$ are explicitly decomposed by powers of ϖ below.

Fast-Fast Block (M_{11}): From Natural Dissipation, the fast–fast transport remainder, and Reverse Perturbation.

$$\begin{aligned} M_{11} &= \big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big) - \alpha k_{\mathrm{reg}} - \alpha\eta k_P(\varpi) \\ &= \gamma\lambda_R - \eta b_{2,\mathrm{eff}} - \alpha k_{\mathrm{reg}} - \alpha\eta(b_1+b_2)k_{\mathcal{A}}\,\varpi^{-1/2}. \end{aligned} \tag{171}$$

Slow-Slow Block (M_{22}): From Main Coercivity, Natural Dissipation (gain), and both Perturbations.

$$\begin{aligned} M_{22} &= \alpha\frac{\lambda(Q_O)}{\varpi+\lambda(Q_O)} - \eta\big(b_2+\tfrac{a_2}{2}\big) - \alpha\eta\,\tilde{k}_P(\varpi) \\ &= \alpha\frac{\lambda(Q_O)}{\varpi+\lambda(Q_O)} - \eta\big(b_2+\tfrac{a_2}{2}\big) - \alpha\eta\left[\Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\varpi^{-1/2}\right]. \end{aligned} \tag{172}$$

Cross Block (M_{12}): From Mixing, Transport Cross terms, and Forward Perturbation.

$$\begin{aligned} M_{12} &= -\frac{1}{2}\alpha\left[\gamma\frac{\sqrt{M_R}}{2}\,\varpi^{-1/2} + \kappa_{\mathrm{ell}} + \eta\big(k_P(\varpi)+\tilde{k}_P(\varpi)\big)\right] \\ &= -\frac{1}{2}\alpha\left[\kappa_{\mathrm{ell}} + \left(\gamma\frac{\sqrt{M_R}}{2} + \eta\Big(2(b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\right)\varpi^{-1/2}\right]. \end{aligned} \tag{173}$$

This explicit form allows for direct asymptotic analysis by choosing $\varpi \sim \lambda(Q_O)$, confirming the stability for small η.

D.4. **Optimization of the Decay Rate.** We work in the unperturbed regime $\eta = 0$ to identify the dominant hypocoercive rate. Let $\mathbf{M}_0(\alpha)$ denote the dissipation matrix at $\eta = 0$. Since $\mathbf{M}_0$ is 2×2 symmetric, its smallest eigenvalue satisfies the determinant–trace bound

$$\lambda_{\min}(\mathbf{M}_0) \ge \frac{\det(\mathbf{M}_0)}{\mathrm{Tr}(\mathbf{M}_0)}. \tag{174}$$

Choice of resolvent shift. The slow-coercivity contribution in M_{22}^0 is $\alpha\lambda(Q_O)/(\varpi+\lambda(Q_O))$, while the cross term $|M_{12}^0|$ contains $\gamma\Theta\varpi^{-1/2}$. A leading-order balance (maximising $\varpi(\varpi+\lambda(Q_O))^{-2}$ in the large-γ rate) motivates the gap-adapted choice

$$\varpi_* := \lambda(Q_O). \tag{175}$$

We do not claim here a separate global optimisation theorem over all admissible shifts; rather, ϖ_* is the canonical gap-scale choice that balances the slow coercive block against the cross term inside the determinant–trace estimate used in this paper. At this value, $\lambda(Q_O)/(\varpi_*+\lambda(Q_O)) = \frac{1}{2}$. Using the constants $\Theta, K_{\mathrm{geom}}, Z(\gamma)$ from (40)–(41), the unperturbed matrix entries at $\varpi = \varpi_*$ are:

$$M_{11}^0 = \gamma\lambda_R - \alpha k_{\mathrm{reg}}, \tag{176}$$

$$M_{22}^0 = \tfrac{1}{2}\,\alpha, \tag{177}$$

$$M_{12}^0 = -\tfrac{1}{2}\,\alpha\,Z(\gamma). \tag{178}$$

Optimal mixing weight. The determinant $D(\alpha) := M_{11}^0 M_{22}^0 - (M_{12}^0)^2$ is a concave quadratic in α:

$$D(\alpha) = \tfrac{1}{2}\gamma\lambda_R\,\alpha - \big(\tfrac{1}{2}k_{\rm reg} + \tfrac{1}{4}Z(\gamma)^2\big)\alpha^2. \tag{179}$$

It is maximised at

$$\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\rm reg}}, \tag{180}$$

in agreement with (42), giving

$$D(\alpha_*) = \frac{\frac{1}{4}(\gamma\lambda_R)^2}{Z(\gamma)^2 + 2k_{\rm reg}}. \tag{181}$$

Trace upper bound and rate estimate. The exact trace is $\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} = (\gamma\lambda_R - \alpha_* k_{\rm reg}) + \frac{1}{2}\alpha_* = \gamma\lambda_R - \alpha_*(k_{\rm reg} - \frac{1}{2})$. Since $k_{\rm reg} = \sqrt{M_R/\lambda_R} \geq 1$ (the flux conditioning bound gives $M_R \geq \lambda_R$), we have $k_{\rm reg} - \frac{1}{2} \geq \frac{1}{2} > 0$, hence

$$\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} \leq \gamma\lambda_R. \tag{182}$$

Applying (174) yields

$$\lambda_{\min}(\mathbf{M}_0) \geq \frac{D(\alpha_*)}{\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*}} \geq \frac{\frac{1}{4}\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\rm reg}} =: \Lambda_{\rm hypo}(\gamma), \tag{183}$$

consistent with definition (43). This provides a rigorous lower bound on the unperturbed decay rate for all $\gamma > 0$.

Asymptotic scaling analysis. Equation (183) reveals the precise scaling behavior across regimes without relying on ad-hoc approximations.
Overdamped Limit ($\gamma \to \infty$). The denominator is dominated by the $\mathcal{O}(\gamma^2)$ term, giving

$$\lim_{\gamma\to\infty} \gamma\Lambda_{\rm hypo}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\lambda(Q_O)}{\Theta^2} = \lambda(Q_O)\,\frac{\lambda_R}{M_R}. \tag{184}$$

Thus, $\Lambda_{\rm hypo}(\gamma) = \lambda(Q_O)\,\frac{\lambda_R}{M_R}\,\gamma^{-1} + \mathcal{O}(\gamma^{-2})$. In particular, $\Lambda_{\rm hypo}(\gamma) \asymp \lambda(Q_O)\,\gamma^{-1}$.
Underdamped/Intermediate Regime. For small or order-one γ, the denominator is controlled by the γ-independent structural constants $K_{\rm geom}$ and $2k_{\rm reg}$. These arise from the slow–fast coupling through $\mathcal{V}_{\rm skew}$ and prevent the estimate from degenerating even though $\mathcal{R}$ does not dissipate on $\mathcal{H}_S$ directly. The transport mixes slow directions into fast dissipative directions, yielding $\Lambda_{\rm hypo}(\gamma) \asymp \gamma$ for small γ.

We now establish the stability condition by computing the operator norm of the perturbation matrix $\mathbf{E}(\gamma)$ as a function of γ, tracking the dependence on the macroscopic spectral gap $\varpi_* = \lambda(Q_O)$.

Explicit norm of the perturbation matrix. Recalling that $\mathbf{M} = \mathbf{M}_0 - \eta\mathbf{E}$, the perturbation matrix $\mathbf{E}$ has entries read off from the full $\mathbf{M}$ computation:

$$E_{11} = b_{2,\rm eff} + \alpha_*\,k_P(\varpi_*), \tag{185}$$

$$E_{22} = \big(b_2 + \tfrac{a_2}{2}\big) + \alpha_*\,\tilde{k}_P(\varpi_*), \tag{186}$$

$$E_{12} = \frac{1}{2}\,\alpha_*\,\big(k_P(\varpi_*) + \tilde{k}_P(\varpi_*)\big). \tag{187}$$

Here E_{11} contains the full effective coefficient $b_{2,\rm eff} = b_2 + \frac{a_2}{2}(1+\lambda_R)$, while E_{22} contains only the slow-mode coefficient $b_2 + \frac{a_2}{2}$; the extra $\frac{a_2\lambda_R}{2}$ absorbed into $b_{2,\rm eff}$ arises in the fast block from the

bound $\|(-\mathcal{R})^{1/2}f\|^2 \geq \lambda_R\|f_F\|^2$ applied to the symmetric perturbation in $\mathcal{D}_I$ (see (135)). Since $\mathbf{E}$ is symmetric, we bound its spectral norm by the maximal absolute row sum (Gershgorin):

$$\|\mathbf{E}(\gamma)\| \leq \max\{E_{11} + |E_{12}|,\, E_{22} + |E_{12}|\}. \tag{188}$$

Using $b_{2,\text{eff}} \geq b_2 + \frac{a_2}{2}$ and $k_P(\varpi_*) \leq \tilde{k}_P(\varpi_*)$, both row sums admit the same uniform upper bound:

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\left[\tfrac{3}{2}\,\tilde{k}_P(\varpi_*) + \tfrac{1}{2}\,\tilde{k}_P(\varpi_*)\right] = b_{2,\text{eff}} + 2\alpha_*(\gamma)\,\tilde{k}_P(\varpi_*). \tag{189}$$

Recall from Lemma 4.6 that the bounds k_P and $\tilde{k}_P$ depend on $\varpi_*^{-1/2}$. We define the *strictly constant* structural coefficients (independent of ϖ and γ):

$$C_{\text{sing}} := 2(b_1 + b_2)k_{\mathcal{A}} + (a_1 + a_2). \tag{190}$$

Indeed,

$$2\tilde{k}_P(\varpi_*) = 2\Big((b_1 + b_2)k_{\mathcal{A}} + \frac{a_1 + a_2}{2}\Big)\varpi_*^{-1/2} = C_{\text{sing}}\,\varpi_*^{-1/2},$$

so the bound (189) becomes

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\,C_{\text{sing}}\,\varpi_*^{-1/2}. \tag{191}$$

Substituting $\alpha_*(\gamma)$ from (42), the bound becomes

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2} = B_{\text{pert}}(\gamma), \tag{192}$$

consistent with definition (44).

Friction-dependent threshold. The stability condition $\lambda_{\min}(\mathbf{M}) > 0$ is ensured by $\eta\|\mathbf{E}(\gamma)\| \leq \frac{1}{2}\Lambda_{\text{hypo}}(\gamma)$, which yields the exact matrix threshold

$$\eta^\sharp(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2\|\mathbf{E}(\gamma)\|}. \tag{193}$$

Since $\|\mathbf{E}(\gamma)\| \leq B_{\text{pert}}(\gamma)$, the explicit certified threshold used in Theorem 5.1 is

$$\eta_{\text{mat}}(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2B_{\text{pert}}(\gamma)} \leq \eta^\sharp(\gamma). \tag{194}$$

Together with the energy-admissibility condition $\eta \leq \eta_{\text{sym}}(\gamma)$ coming from (135), this gives the theorem threshold

$$\eta^*(\gamma) := \min\{\eta_{\text{sym}}(\gamma), \eta_{\text{mat}}(\gamma)\}. \tag{195}$$

Substituting the explicit forms of Λ_{hypo} (43) and B_{pert} (44) yields

$$\eta_{\text{mat}}(\gamma) = \frac{\frac{1}{4}\,\gamma\lambda_R}{\big(Z(\gamma)^2 + 2k_{\text{reg}}\big)\big(2b_{2,\text{eff}} + 2\alpha_*(\gamma)\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2}\big)}. \tag{196}$$

Asymptotic analysis of the certified threshold. We analyze first the scaling of $\eta_{\text{mat}}(\gamma)$ in the limiting regimes. Note that $Z(\gamma) \asymp \gamma$ as $\gamma \to \infty$.

Overdamped Limit ($\gamma \to \infty$). The numerator scales as $\mathcal{O}(\gamma)$. In the denominator, the term $2b_{2,\text{eff}}Z(\gamma)^2$ scales as $\mathcal{O}(\gamma^2)$, while the skew-term scales as $\mathcal{O}(\gamma)$. If $b_{2,\text{eff}} > 0$ (symmetric perturbation present), the $\mathcal{O}(\gamma^2)$ term dominates and $\eta_{\text{mat}}(\gamma) \asymp \gamma^{-1}$. If $b_{2,\text{eff}} = 0$ (purely skew perturbation, i.e. $b_2 = a_2 = 0$), the denominator is dominated by the skew term $\mathcal{O}(\gamma)$ and $\eta_{\text{mat}}(\gamma) \asymp \mathcal{O}(1)$.

Optimal Lift ($\gamma \sim \sqrt{\lambda(Q_O)}$). At the optimal friction γ_{opt}, determined in the optimization step below, we have $Z(\gamma_{\text{opt}}) = \mathcal{O}(1)$. Consider the singular term in the denominator:

$$\gamma_{\text{opt}} \cdot C_{\text{sing}}\lambda(Q_O)^{-1/2} \asymp \sqrt{\lambda(Q_O)} \cdot \lambda(Q_O)^{-1/2} \asymp \mathcal{O}(1). \tag{197}$$

The singularity thus cancels, and at the optimal friction the matrix-controlled threshold is well-behaved:

$$\eta_{\mathrm{mat}}(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{198}$$

Since $\eta_{\mathrm{sym}}(\gamma_{\mathrm{opt}}) = +\infty$ when $a_2 = 0$ and $\eta_{\mathrm{sym}}(\gamma_{\mathrm{opt}}) = 2\gamma_{\mathrm{opt}}/a_2 \asymp \sqrt{\lambda(Q_O)}$ when $a_2 > 0$, the overall threshold $\eta^*(\gamma_{\mathrm{opt}}) = \min\{\eta_{\mathrm{sym}}(\gamma_{\mathrm{opt}}), \eta_{\mathrm{mat}}(\gamma_{\mathrm{opt}})\}$ has the same $\sqrt{\lambda(Q_O)}$ scaling.

From coercivity to global decay. Having established the positivity of the dissipation matrix, we convert the algebraic estimate into a global semigroup decay bound. This step involves explicit algebraic substitution to derive the final decay rate, ensuring that the singular scaling of the perturbation is correctly propagated.

Uniform norm equivalence. Recall the distortion parameter

$$\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|. \tag{199}$$

Using $\|\mathcal{A}_{\varpi_*}\| \le k_{\mathcal{A}}\, \varpi_*^{-1/2}$ with $k_{\mathcal{A}} = (2\sqrt{\lambda_R})^{-1}$ and the explicit weight $\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2+2k_{\mathrm{reg}}}$, we obtain

$$\delta_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\mathrm{reg}}}\,\frac{1}{\sqrt{\lambda_R\,\varpi_*}} = \frac{\gamma\sqrt{\lambda_R}\,\varpi_*^{-1/2}}{\big(\gamma\Theta\varpi_*^{-1/2} + K_{\mathrm{geom}}\big)^2 + 2k_{\mathrm{reg}}}. \tag{200}$$

Setting $x := \gamma\Theta\varpi_*^{-1/2}$ gives

$$\delta_*(\gamma) = \frac{\sqrt{\lambda_R}}{\Theta}\cdot\frac{x}{(x+K_{\mathrm{geom}})^2 + 2k_{\mathrm{reg}}}. \tag{201}$$

The elementary optimisation

$$\sup_{x>0}\frac{x}{(x+K_{\mathrm{geom}})^2 + 2k_{\mathrm{reg}}} = \frac{1}{2\big(K_{\mathrm{geom}} + \sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}}\big)} \tag{202}$$

together with $M_R \ge \lambda_R$ and $\sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}} \ge \sqrt{2k_{\mathrm{reg}}} \ge \sqrt{2}$ (the latter since $k_{\mathrm{reg}} = \sqrt{M_R/\lambda_R} \ge 1$) therefore yields

$$\delta_*(\gamma) \le \frac{\sqrt{\lambda_R}}{\sqrt{M_R}\big(K_{\mathrm{geom}} + \sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}}\big)} \le 2^{-1/2}, \tag{203}$$

and therefore

$$\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\| = \frac{\delta_*(\gamma)}{2} \le 2^{-3/2} < \frac{1}{2}. \tag{204}$$

Thus Proposition 4.8 applies with $(\alpha, \varpi) = (\alpha_*, \varpi_*)$. Hence the norm-equivalence prefactor satisfies

$$C_*(\gamma) := \sqrt{\frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}} \le 1 + \sqrt{2}. \tag{205}$$

Thus, for all f,

$$\frac{1}{2}(1-\delta_*(\gamma))\|f\|^2 \le \mathcal{E}_{\alpha_*,\varpi_*}(f) \le \frac{1}{2}(1+\delta_*(\gamma))\|f\|^2. \tag{206}$$

Explicit calculation of the decay rate. Set

$$c(\gamma,\eta) := \Lambda_{\mathrm{hypo}}(\gamma) - \eta\,\|\mathbf{E}(\gamma)\|. \tag{207}$$

The algebraic dissipation estimate reads

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le -c(\gamma,\eta)\,\|f_t\|^2. \tag{208}$$

Next we use the norm equivalence from Proposition 4.8 in the convenient form

$$\frac{1}{2}(1-\delta_*(\gamma))\,\|f\|^2 \le \mathcal{E}_{\alpha_*,\varpi_*}(f) \le \frac{1}{2}(1+\delta_*(\gamma))\,\|f\|^2, \tag{209}$$

where $\delta_*(\gamma) := 2\alpha_*(\gamma)\,\|\mathcal{A}_{\varpi_*}\| \in (0,1)$. The upper bound implies

$$\|f\|^2 \ge \frac{2}{1+\delta_*(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(f). \tag{210}$$

Substituting this into the dissipation inequality gives a closed differential inequality for the Lyapunov functional:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le -\frac{2c(\gamma,\eta)}{1+\delta_*(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t). \tag{211}$$

By Grönwall's inequality,

$$\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le \exp\Big(-\frac{2c(\gamma,\eta)}{1+\delta_*(\gamma)}\,t\Big)\,\mathcal{E}_{\alpha_*,\varpi_*}(f_0). \tag{212}$$

Finally, converting back to the L^2 norm using the two-sided equivalence yields

$$\|f_t\|^2 \le \frac{2}{1-\delta_*(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le \frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}\,e^{-\frac{2c(\gamma,\eta)}{1+\delta_*(\gamma)}t}\,\|f_0\|^2. \tag{213}$$

Therefore

$$\|f_t\| \le \sqrt{\frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}}\,e^{-\lambda_\eta t}\,\|f_0\|, \qquad \lambda_\eta := \frac{c(\gamma,\eta)}{1+\delta_*(\gamma)}. \tag{214}$$

We now substitute the explicit expressions derived in the previous steps. For the unperturbed rate:

$$\Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\gamma}{Z(\gamma)^2+2k_{\text{reg}}}, \quad \text{with } Z(\gamma) = \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}. \tag{215}$$

For the perturbation norm, we use the sharp bound (191) at $\varpi_* = \lambda(Q_O)$ to handle the macroscopic singularity:

$$\|\mathbf{E}(\gamma)\| \le b_{2,\text{eff}} + \underbrace{\frac{\gamma\lambda_R}{Z(\gamma)^2+2k_{\text{reg}}}}_{\alpha_*(\gamma)}\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2}. \tag{216}$$

Substituting these into the expression for λ_η and using the explicit distortion bound on $\delta_*(\gamma)$:

$$\lambda_\eta \ge \frac{\frac{1}{4}\lambda_R\,\gamma}{(1+\delta_*(\gamma))(Z(\gamma)^2+2k_{\text{reg}})} - \frac{\eta}{1+\delta_*(\gamma)}\left(b_{2,\text{eff}} + \frac{\gamma\lambda_R}{Z(\gamma)^2+2k_{\text{reg}}}\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2}\right). \tag{217}$$

Equivalently,

$$\lambda_\eta \ge \frac{\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)}{1+\delta_*(\gamma)} \ge (2-\sqrt{2})\big(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)\big), \tag{218}$$

using $1+\delta_*(\gamma) \le 1+2^{-1/2}$, i.e. $(1+\delta_*(\gamma))^{-1} \ge (1+2^{-1/2})^{-1} = 2-\sqrt{2}$. This is the rigorous lower bound. The potentially singular factor $C_{\text{sing}}\lambda(Q_O)^{-1/2}$ is multiplied by the mixing weight $\alpha_*(\gamma) = \gamma\lambda_R/(Z(\gamma)^2+2k_{\text{reg}})$, which scales as $\sqrt{\lambda(Q_O)}$ at the optimal lift, ensuring the perturbation contribution remains bounded.

Finally, converting back to the L^2 norm via Grönwall's lemma yields the global estimate:

$$\|e^{t\mathcal{L}}f\|_{\mathcal{H}} \le C_*(\gamma)\, e^{-\lambda_\eta t}\|f\|_{\mathcal{H}} \le (1+\sqrt{2})\, e^{-\lambda_\eta t}\|f\|_{\mathcal{H}}. \tag{219}$$

This completes the proof.

D.5. **Optimization over Friction γ: Kinetic Acceleration.** We determine $\gamma_{\rm opt}$ maximising $\Lambda_{\rm hypo}(\gamma)$. From (183), setting $x := \Theta\lambda(Q_O)^{-1/2}\gamma$, the rate function reads

$$\Lambda_{\rm hypo}(\gamma) = \frac{\frac{1}{4}\lambda_R}{\Theta}\sqrt{\lambda(Q_O)}\cdot\frac{x}{(x+K_{\rm geom})^2+2k_{\rm reg}}. \tag{220}$$

Optimal friction. The function $x \mapsto x/[(x+K_{\rm geom})^2+C_0]$ with $C_0 := 2k_{\rm reg}$ is maximised when $(x+K_{\rm geom})^2+C_0 = 2x(x+K_{\rm geom})$, equivalently $x^2 = K_{\rm geom}^2 + C_0$. Translating back:

$$\gamma_{\rm opt} = \frac{\sqrt{K_{\rm geom}^2+2k_{\rm reg}}}{\Theta}\sqrt{\lambda(Q_O)} =: C_{\rm lift}\sqrt{\lambda(Q_O)}, \tag{221}$$

where $C_{\rm lift} := \Theta^{-1}\sqrt{K_{\rm geom}^2+2k_{\rm reg}}$.

Maximal rate. At $\gamma = \gamma_{\rm opt}$ one has $x_* := \Theta\lambda(Q_O)^{-1/2}\gamma_{\rm opt} = \sqrt{K_{\rm geom}^2+C_0}$. Substituting into the denominator: $(x_*+K_{\rm geom})^2+C_0 = 2(K_{\rm geom}^2+C_0+K_{\rm geom}x_*)$. Hence

$$\Lambda_{\max} := \Lambda_{\rm hypo}(\gamma_{\rm opt}) = \frac{\frac{1}{4}\lambda_R}{2\Theta(x_*+K_{\rm geom})}\sqrt{\lambda(Q_O)} \asymp \sqrt{\lambda(Q_O)}. \tag{222}$$

This confirms the kinetic acceleration: $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$, compared to the overdamped ceiling $\Lambda_{\rm hypo}(\gamma) \asymp \lambda(Q_O)/\gamma$ as $\gamma \to \infty$.

Certified threshold at optimal lift. The admissible perturbation range certified by Theorem 5.1 is determined as follows. From (193), the exact matrix threshold depends on $\|\mathbf{E}(\gamma)\|$, while the theorem uses the explicit upper bound $B_{\rm pert}(\gamma)$. The dominant term involves the mixing weight $\alpha_*(\gamma)$. Evaluating at $\gamma_{\rm opt}$:

$$\alpha_*(\gamma_{\rm opt}) = \frac{\gamma_{\rm opt}\lambda_R}{Z(\gamma_{\rm opt})^2+2k_{\rm reg}}. \tag{223}$$

Using $\gamma_{\rm opt} = C_{\rm lift}\sqrt{\lambda(Q_O)}$ and $Z(\gamma_{\rm opt}) = \sqrt{K_{\rm geom}^2+2k_{\rm reg}} + K_{\rm geom} = \mathcal{O}(1)$, the numerator of $\alpha_*(\gamma_{\rm opt})$ is $\gamma_{\rm opt}\lambda_R \asymp \sqrt{\lambda(Q_O)}$ and the denominator $Z(\gamma_{\rm opt})^2+2k_{\rm reg} = \mathcal{O}(1)$ (independent of $\lambda(Q_O)$). Hence $\alpha_*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}$, so $\alpha_*(\gamma_{\rm opt})\cdot\lambda(Q_O)^{-1/2} = \mathcal{O}(1)$. Applying (191):

$$\|\mathbf{E}(\gamma_{\rm opt})\| \le b_{2,\rm eff} + \alpha_*(\gamma_{\rm opt})\, C_{\rm sing}\, \lambda(Q_O)^{-1/2} = \mathcal{O}(1). \tag{224}$$

Since $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$ and $B_{\rm pert}(\gamma_{\rm opt}) = \mathcal{O}(1)$, the matrix-controlled threshold satisfies

$$\eta_{\rm mat}(\gamma_{\rm opt}) = \frac{\Lambda_{\max}}{2B_{\rm pert}(\gamma_{\rm opt})} \asymp \sqrt{\lambda(Q_O)}. \tag{225}$$

Together with $\eta_{\rm sym}(\gamma_{\rm opt}) = +\infty$ when $a_2 = 0$ and $\eta_{\rm sym}(\gamma_{\rm opt}) = 2\gamma_{\rm opt}/a_2 \asymp \sqrt{\lambda(Q_O)}$ when $a_2 > 0$, this yields

$$\eta^*(\gamma_{\rm opt}) = \min\{\eta_{\rm sym}(\gamma_{\rm opt}), \eta_{\rm mat}(\gamma_{\rm opt})\} \asymp \sqrt{\lambda(Q_O)}. \tag{226}$$

This confirms that the accelerated regime retains an $\mathcal{O}(\sqrt{\lambda(Q_O)})$ admissible perturbation range.

Appendix E. Upper bound via singular value gap

This appendix proves Theorem 5.4.

E.1. **Exponential decay implies a singular value-gap bound.** For a (possibly unbounded) closed operator T on $\mathcal{H}_0$, define its injectivity modulus

$$s(T) := \inf_{f \in \mathrm{dom}(T) \cap \mathcal{H}_0 \setminus \{0\}} \frac{\|Tf\|}{\|f\|}. \tag{227}$$

Lemma E.1 (From exponential decay to a singular value-gap bound)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$. *Assume moreover that* $e^{t\mathcal{L}}$ *is a contraction semigroup on* $\mathcal{H}_0$*, i.e.* $\|e^{t\mathcal{L}} f\| \le \|f\|$ *for all* $t \ge 0$ *and* $f \in \mathcal{H}_0$ *(equivalently,* $\mathcal{L}$ *is dissipative on* $\mathcal{H}_0$*:* $\mathrm{Re}\langle \mathcal{L}f, f\rangle \le 0$*). If there exist constants* $C \ge 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}} f\| \le C\, e^{-\nu t}\, \|f\|, \qquad \forall t \ge 0,\ \forall f \in \mathcal{H}_0, \tag{228}$$

then 0 *belongs to the resolvent set of* $\mathcal{L}|_{\mathcal{H}_0}$ *and*

$$\nu \le (1 + \log C)\, s(\mathcal{L}). \tag{229}$$

Proof. Set $t_* := (\log C)/\nu \ge 0$ (the time at which $Ce^{-\nu t}$ crosses 1). For $\lambda > 0$, define the resolvent regularization

$$u_\lambda := -\int_0^\infty e^{-\lambda t} e^{t\mathcal{L}} f\, dt. \tag{230}$$

This Bochner integral is absolutely convergent because $e^{-\lambda t} \le 1$ and the semigroup is bounded by the piecewise integrable envelope used below. On $[0, t_*]$ use the contraction bound $\|e^{t\mathcal{L}} f\| \le \|f\|$; on $[t_*, \infty)$ use the exponential decay bound $\|e^{t\mathcal{L}} f\| \le Ce^{-\nu t}\|f\|$. Combining these estimates and the inequality $e^{-\lambda t} \le 1$ gives

$$\|u_\lambda\| \le \int_0^{t_*} \|f\|\, dt + \int_{t_*}^\infty Ce^{-\nu t}\|f\|\, dt = t_*\, \|f\| + \frac{C}{\nu} e^{-\nu t_*}\, \|f\| = \frac{\log C}{\nu}\, \|f\| + \frac{1}{\nu}\, \|f\| = \frac{1 + \log C}{\nu}\, \|f\|, \tag{231}$$

where we used $Ce^{-\nu t_*} = Ce^{-\log C} = 1$. By the Hille–Yosida resolvent formula, $u_\lambda \in \mathrm{dom}(\mathcal{L})$ and

$$(\lambda - \mathcal{L})(-u_\lambda) = f, \qquad \text{equivalently} \qquad \mathcal{L}u_\lambda = f + \lambda u_\lambda. \tag{232}$$

The family $(u_\lambda)_{\lambda > 0}$ is uniformly bounded in $\mathcal{H}_0$ by the estimate above and converges, as $\lambda \downarrow 0$, to

$$u := -\int_0^\infty e^{t\mathcal{L}} f\, dt \tag{233}$$

by dominated convergence. Since $\lambda u_\lambda \to 0$ in $\mathcal{H}_0$ and $\mathcal{L}$ is closed, the relation $\mathcal{L}u_\lambda = f + \lambda u_\lambda$ passes to the limit and gives $u \in \mathrm{dom}(\mathcal{L})$ together with $\mathcal{L}u = f$. Thus $(\mathcal{L}|_{\mathcal{H}_0})^{-1}$ exists and satisfies $\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\| \le (1 + \log C)/\nu$, hence $s(\mathcal{L}) = \|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1} \ge \nu/(1 + \log C)$, i.e. $\nu \le (1 + \log C)\, s(\mathcal{L})$. □

Lemma E.2 (Relaxation time is bounded below by the singular value gap)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$. *Assume moreover that* $e^{t\mathcal{L}}$ *is a contraction semigroup on* $\mathcal{H}_0$*, i.e.* $\|e^{t\mathcal{L}} f\| \le \|f\|$ *for all* $t \ge 0$ *and* $f \in \mathcal{H}_0$*. Assume there exist constants* $C \ge 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}} f\| \le C\, e^{-\nu t}\, \|f\|, \qquad \forall t \ge 0,\ \forall f \in \mathcal{H}_0. \tag{234}$$

Define the L^2 *relaxation time*

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{ t \ge 0\,;\ \|e^{t\mathcal{L}} f\| \le e^{-1}\, \|f\| \text{ for all } f \in \mathcal{H}_0 \Big\}. \tag{235}$$

Then

$$t_{\mathrm{rel}}(\mathcal{L}) \ge \frac{1}{2\, s(\mathcal{L})}. \tag{236}$$

This L^2 *bound is a continuous-time analogue of discrete-time relaxation estimates based on singular values for non-reversible Markov chains; compare* [6].

Proof. Let $\tau > t_{\mathrm{rel}}(\mathcal{L})$ be such that

$$\|e^{\tau\mathcal{L}} f\| \le e^{-1}\|f\|, \qquad \forall f \in \mathcal{H}_0. \tag{237}$$

Such a τ exists by the definition of the infimum. For any $t \ge 0$ write $t = n\tau + r$ with $n \in \mathbb{N}_0$ and $r \in [0, \tau)$. The contraction property and the semigroup identity yield

$$\|e^{t\mathcal{L}}\| \le \|e^{r\mathcal{L}}\|\, \|e^{\tau\mathcal{L}}\|^n \le e^{-n}. \tag{238}$$

Since $n \ge t/\tau - 1$, this implies

$$\|e^{t\mathcal{L}}\| \le e\, e^{-t/\tau}. \tag{239}$$

We may therefore apply Lemma E.1 with $C = e$ and $\nu = \tau^{-1}$ to obtain

$$\tau^{-1} \le (1 + \log e)\, s(\mathcal{L}) = 2\, s(\mathcal{L}). \tag{240}$$

Hence $\tau \ge (2s(\mathcal{L}))^{-1}$. Letting $\tau \downarrow t_{\mathrm{rel}}(\mathcal{L})$ gives the stated bound. □

E.2. **Test function and completion of the proof.**

Proof of Theorem 5.4. Let $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0$ be a normalized eigenfunction of Q_O with

$$Q_O\phi = -\lambda(Q_O)\, \phi, \qquad \|\phi\| = 1. \tag{241}$$

Define the first-order corrector

$$\psi := \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi \in \mathcal{H}_F, \tag{242}$$

and the test function $f_\gamma := \phi - \gamma^{-1}\psi$. Since $\phi \in \mathcal{H}_S$ and $\psi \in \mathcal{H}_F$ we have $\langle \phi, \psi\rangle = 0$ and therefore

$$\|f_\gamma\|^2 = \|\phi\|^2 + \gamma^{-2}\|\psi\|^2 \ge 1. \tag{243}$$

Using $\mathcal{R}\phi = 0$ and $\mathcal{R}\mathcal{R}^+ = P_F$ on $\mathcal{H}$, together with centered coupling $\mathcal{V}_{\mathrm{skew}}\phi \in \mathcal{H}_F$, we obtain

$$\mathcal{R}\psi = \mathcal{R}\mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi = P_F\mathcal{V}_{\mathrm{skew}}\phi = \mathcal{V}_{\mathrm{skew}}\phi. \tag{244}$$

Consequently,

$$\mathcal{L}f_\gamma = \big(\gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}\big)\big(\phi - \gamma^{-1}\psi\big) \tag{245}$$

$$= \underbrace{\mathcal{V}_{\mathrm{skew}}\phi - \mathcal{R}\psi}_{=0} - \gamma^{-1}\mathcal{V}_{\mathrm{skew}}\psi \;+\; \eta\mathcal{V}_{\mathrm{pert}}f_\gamma. \tag{246}$$

We decompose the remaining transport term into slow/fast components:

$$\mathcal{V}_{\mathrm{skew}}\psi = P_S\mathcal{V}_{\mathrm{skew}}\psi + P_F\mathcal{V}_{\mathrm{skew}}\psi. \tag{247}$$

For the slow part, using $Q_O := -P_S\mathcal{V}_{\mathrm{skew}}\mathcal{R}^+\mathcal{V}_{\mathrm{skew}}P_S$ and $P_S\phi = \phi$, we have

$$P_S\mathcal{V}_{\mathrm{skew}}\psi = P_S\mathcal{V}_{\mathrm{skew}}\mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi = -Q_O\phi = \lambda(Q_O)\, \phi. \tag{248}$$

For the fast part, apply the flux regularity estimate (23) with the slow input ϕ. Writing $g_\phi := P_F\mathcal{V}_{\mathrm{skew}}P_S\phi = \mathcal{V}_{\mathrm{skew}}\phi$ and noting that $(-\mathcal{R}^+)g_\phi = -\psi$ on $\mathcal{H}_F$, we obtain

$$\|P_F\mathcal{V}_{\mathrm{skew}}\psi\| = \|P_F\mathcal{V}_{\mathrm{skew}}P_F\psi\| = \|P_F\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)g_\phi\| \le \kappa_{\mathrm{ell}}\, \|(-Q_O)\phi\| = \kappa_{\mathrm{ell}}\, \lambda(Q_O). \tag{249}$$

Combining the slow and fast contributions yields

$$\|\mathcal{V}_{\mathrm{skew}}\psi\| \le (1 + \kappa_{\mathrm{ell}})\, \lambda(Q_O). \tag{250}$$

Hence

$$\|\mathcal{L}f_\gamma\| \le \frac{(1 + \kappa_{\mathrm{ell}})\, \lambda(Q_O)}{\gamma} + \eta\, \|\mathcal{V}_{\mathrm{pert}}f_\gamma\|. \tag{251}$$

To bound the perturbation, use Assumption 4.1.4 to estimate

$$\|\mathcal{V}_{\rm pert} f\| \le \|\mathcal{K}_{\rm pert} f\| + \|\mathcal{S}_{\rm pert} f\| \tag{252}$$

$$\le (a_1 + a_2)\, \|(-\mathcal{R})^{1/2} f\| + (b_1 + b_2)\, \|f\|, \qquad f \in \mathrm{dom}((-\mathcal{R})^{1/2}). \tag{253}$$

Since $\phi \in \ker(\mathcal{R})$ and $f_\gamma = \phi - \gamma^{-1}\psi$, we have

$$\|(-\mathcal{R})^{1/2} f_\gamma\| = \gamma^{-1}\, \|(-\mathcal{R})^{1/2}\psi\|. \tag{254}$$

Finally, using $\psi = \mathcal{R}^+ \mathcal{V}_{\rm skew}\phi$ and the definition of Q_O,

$$\|(-\mathcal{R})^{1/2}\psi\|^2 = \langle(-\mathcal{R})\psi, \psi\rangle = \langle \mathcal{V}_{\rm skew}\phi, (-\mathcal{R}^+)\mathcal{V}_{\rm skew}\phi\rangle = -\langle Q_O\phi, \phi\rangle = \lambda(Q_O). \tag{255}$$

Therefore $\|(-\mathcal{R})^{1/2} f_\gamma\| = \sqrt{\lambda(Q_O)}/\gamma$, and

$$\|\mathcal{V}_{\rm pert} f_\gamma\| \le (b_1 + b_2)\, \|f_\gamma\| + (a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{256}$$

Dividing by $\|f_\gamma\| \ge 1$ gives

$$\frac{\|\mathcal{L} f_\gamma\|}{\|f_\gamma\|} \le \frac{(1 + \kappa_{\rm ell})\, \lambda(Q_O)}{\gamma} + \eta\, (b_1 + b_2) + \eta\, (a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{257}$$

By definition of $s(\mathcal{L})$, this implies $s(\mathcal{L}) \le \|\mathcal{L} f_\gamma\|/\|f_\gamma\|$. Combining with Lemma E.1 yields the claimed bound on ν. □

Appendix F. Proof of Proposition 6.2

We verify Assumptions 4.1.1–4 and 4.3 for a uniformly convex potential $U \in C^\infty(\mathbb{R}^d)$ with $\nabla^2 U \ge m > 0$ and with all derivatives of positive order growing at most polynomially,

$$\mathcal{V}_{\rm pert} f = F(q) \cdot \nabla_p f, \tag{258}$$

with $d \ge 2$, $m \in (0, 1]$, and $F \in C_b^\infty(\mathbb{R}^d; \mathbb{R}^d)$ bounded. We work on the Schwartz core

$$\mathcal{C}_{\rm Lan} := \mathcal{S}(\mathbb{R}^d_q \times \mathbb{R}^d_p) \subset L^2(\mu), \tag{259}$$

where

$$d\mu(q, p) = Z^{-1} e^{-\beta(\frac{1}{2}|p|^2 + U(q))}\, dq\, dp.$$

The only non-quantitative input is the standard semigroup/common-core theory for confining Kramers–Fokker–Planck operators, which we record first.

F.1. **Analytic input: semigroup generation and Schwartz core.**

Proposition F.1 (Semigroup generation and Schwartz core)**.** *Under the hypotheses of Proposition 6.2—in particular, the smooth polynomial-growth assumptions on U from Example 6.2—the closure of $\mathcal{L}|_{\mathcal{C}_{\rm Lan}}$ generates the standard nonequilibrium Langevin semigroup on $L^2(\mu)$. Moreover, $\mathcal{C}_{\rm Lan} = \mathcal{S}(\mathbb{R}^{2d})$ is invariant under $\mathcal{R}$, $\mathcal{V}_{\rm skew}$, $\mathcal{V}_{\rm pert}$, P_S, and P_F; the pseudoinverse $\mathcal{R}^+$ preserves $\mathcal{C}_{\rm Lan}$; for every slow test function $\psi \in \mathcal{C}_{\rm Lan} \cap \mathcal{H}_S = \mathcal{S}(\mathbb{R}^d)$ one has*

$$\mathcal{R}^+ \mathcal{V}_{\rm skew}\psi = -p \cdot \nabla_q \psi \in \mathcal{C}_{\rm Lan}; \tag{260}$$

and $\mathcal{C}_{\rm Lan} \cap \mathcal{H}_S$ is a core for the Friedrichs realization of

$$Q_O = \beta^{-1}\Delta_q - \nabla_q U \cdot \nabla_q \tag{261}$$

on $L^2(\mu_q)$. Consequently Assumption 4.3 holds.

Proof. The semigroup-generation and common-core statements are standard for confining Kramers–Fokker–Planck operators with smooth polynomial coefficients and bounded first-order perturbations in the momentum variable; see [17, Introduction and §§1–2], [10, §2], and [16]. We record only the concrete facts used in this paper.

Because the coefficients of $\mathcal{R}$, $\mathcal{V}_{\text{skew}}$, and $\mathcal{V}_{\text{pert}}$ are smooth and have at most polynomial growth, each of these differential operators maps $\mathcal{C}_{\text{Lan}}$ into itself, and the Gaussian conditional expectation operators P_S and P_F preserve $\mathcal{C}_{\text{Lan}}$ as well. The Ornstein–Uhlenbeck operator $\mathcal{R}$ diagonalizes in the Hermite basis in the p-variable, so $\mathcal{R}^+$ preserves the Schwartz class, and for slow $\psi = \psi(q)$ one obtains $\mathcal{R}^+\mathcal{V}_{\text{skew}}\psi = -p \cdot \nabla_q \psi \in \mathcal{C}_{\text{Lan}}$.

On $L^2(\mu_q)$, the operator Q_O is the weighted Laplacian with uniformly convex potential. Its non-positive Friedrichs realization has form domain $H^1(\mu_q)$, and standard Lax–Milgram/Friedrichs theory yields that $(\varpi - Q_O)^{-1}$ maps $L^2(\mu_q)$ into $\operatorname{dom}(Q_O)$ for every $\varpi > 0$. Since $C_c^\infty(\mathbb{R}^d)$ is a core for this realization, the larger class $\mathcal{S}(\mathbb{R}^d) = \mathcal{C}_{\text{Lan}} \cap \mathcal{H}_S$ is a core as well. These facts give exactly the common-core, mapping, and closure properties required in Assumption 4.3. □

With Proposition F.1 fixed, the remainder of the appendix is purely quantitative: we verify the structural constants in Assumptions 4.1.1–4 and the slow gap in Assumption 4.2. The differential identities below are first checked on $\mathcal{C}_{\text{Lan}}$ and then extended by closure.

F.2. **Microscopic spectral gap, centering, and slow gap.** The fast dissipator

$$\mathcal{R} = -p \cdot \nabla_p + \beta^{-1}\Delta_p \tag{262}$$

acts only on the Gaussian momentum variable and has spectral gap

$$\lambda_R = 1 \tag{263}$$

on $\mathcal{H}_F = \ker(\Pi_p)$. The Hamiltonian part $\mathcal{V}_{\text{skew}} = p \cdot \nabla_q - \nabla_q U \cdot \nabla_p$ is skew-adjoint in $L^2(\mu)$ because the Hamiltonian flow preserves μ. Since $\int p \, d\mu_p = 0$, one has

$$P_S \mathcal{V}_{\text{skew}} P_S = 0. \tag{264}$$

Also $P_S \mathcal{V}_{\text{pert}} P_S = 0$ because $\nabla_p \phi(q) = 0$ for slow functions.

For $\psi = \psi(q)$,

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi = p \cdot \nabla_q \psi. \tag{265}$$

Since $\mathcal{R}(p_i) = -p_i$, one has $(-\mathcal{R}^+) g_\psi = g_\psi$, so the conditioning constant is

$$M_R = 1. \tag{266}$$

The effective slow operator is

$$Q_O \psi = \beta^{-1}\Delta_q \psi - \nabla_q U \cdot \nabla_q \psi. \tag{267}$$

By the Poincaré inequality for the uniformly log-concave measure $\mu_q(dq) \propto e^{-\beta U(q)}\, dq$ with constant m—equivalently, the Brascamp–Lieb/Bakry–Émery spectral-gap estimate associated with $\nabla^2 U \geq m > 0$ [1]—one has

$$-\langle \psi, Q_O \psi \rangle_{L^2(\mu_q)} = \beta^{-1} \|\nabla_q \psi\|^2_{L^2(\mu_q)} \geq m \, \|\psi - \textstyle\int \psi \, d\mu_q\|^2_{L^2(\mu_q)}, \tag{268}$$

so Assumption 4.2 holds with

$$\lambda(Q_O) = m. \tag{269}$$

F.3. **Flux regularity (Assumption 4.1.3).** Define

$$Z_{ij}(p) := p_i p_j - \beta^{-1}\delta_{ij}. \tag{270}$$

Using $(-\mathcal{R}^+)g_\psi = g_\psi$, we compute

$$\begin{aligned}\mathcal{V}_{\text{skew}} g_\psi &= p\cdot\nabla_q(p\cdot\nabla_q\psi) - \nabla_q U\cdot\nabla_p(p\cdot\nabla_q\psi)\\ &= \sum_{i,j=1}^d p_i p_j\,\partial_{q_iq_j}\psi - \nabla_q U\cdot\nabla_q\psi.\end{aligned} \tag{271}$$

The p-average of this expression is

$$P_S\mathcal{V}_{\text{skew}} g_\psi = \beta^{-1}\Delta_q\psi - \nabla_q U\cdot\nabla_q\psi = Q_O\psi, \tag{272}$$

so subtracting the slow part gives

$$P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\psi = \sum_{i,j=1}^d (p_ip_j - \beta^{-1}\delta_{ij})\,\partial_{q_iq_j}\psi = \sum_{i,j=1}^d Z_{ij}(p)\,\partial_{q_iq_j}\psi. \tag{273}$$

The Gaussian fourth-moment identity

$$\mathbb{E}_{\mu_p}[Z_{ij}Z_{kl}] = \beta^{-2}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) \tag{274}$$

therefore gives

$$\|P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\psi\|^2_{L^2(\mu)} = 2\beta^{-2}\,\|\nabla_q^2\psi\|^2_{L^2(\mu_q)}. \tag{275}$$

To control the Hessian, we use the Bochner identity. Set $L := -\beta Q_O = -\Delta_q + \beta\nabla_q U\cdot\nabla_q$, the weighted Laplacian on $L^2(\mu_q)$. For every $\psi\in\mathcal{S}(\mathbb{R}^d)$, one has the commutator

$$[\partial_{q_i}, L] = \beta\,(\partial^2_{q_iq_k}U)\,\partial_{q_k}, \tag{276}$$

obtained from $[\partial_{q_i}, \nabla_q U\cdot\nabla_q] = (\partial^2_{q_iq_k}U)\partial_{q_k}$ and $[\partial_{q_i}, -\Delta_q] = 0$. Since L is self-adjoint in $L^2(\mu_q)$ and $\langle g, Lg\rangle_{L^2(\mu_q)} = \|\nabla_q g\|^2_{L^2(\mu_q)}$,

$$\begin{aligned}\|L\psi\|^2_{L^2(\mu_q)} = \langle\psi, L^2\psi\rangle_{L^2(\mu_q)} &= \sum_{i=1}^d\langle\partial_{q_i}\psi, \partial_{q_i}(L\psi)\rangle_{L^2(\mu_q)}\\ &= \sum_{i=1}^d\langle\partial_{q_i}\psi, L(\partial_{q_i}\psi)\rangle_{L^2(\mu_q)} + \beta\,\langle\nabla_q^2U\,\nabla_q\psi, \nabla_q\psi\rangle_{L^2(\mu_q)}\\ &= \|\nabla_q^2\psi\|^2_{L^2(\mu_q)} + \beta\,\langle\nabla_q^2U\,\nabla_q\psi, \nabla_q\psi\rangle_{L^2(\mu_q)}.\end{aligned} \tag{277}$$

Since U is convex ($\nabla_q^2U\geq 0$), the second term is non-negative, giving the *Bochner inequality*

$$\|\nabla_q^2\psi\|_{L^2(\mu_q)} \leq \|L\psi\|_{L^2(\mu_q)} = \beta\,\|(-Q_O)\psi\|_{L^2(\mu_q)}. \tag{278}$$

Combining with (275) yields

$$\|P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\psi\|_{L^2(\mu)} \leq \sqrt{2}\,\|(-Q_O)\psi\|_{L^2(\mu_q)}. \tag{279}$$

Thus Assumption 4.1.3 holds with

$$\kappa_{\text{ell}} = \sqrt{2}. \tag{280}$$

F.4. **Perturbation bounds (Assumption 4.1.4).** The $L^2(\mu)$-adjoint of $\mathcal{V}_{\text{pert}} = F(q)\cdot\nabla_p$ is

$$\mathcal{V}_{\text{pert}}^\dagger = -F(q)\cdot\nabla_p + \beta F(q)\cdot p, \tag{281}$$

so the skew/symmetric decomposition is

$$\mathcal{K}_{\text{pert}} = F(q)\cdot\nabla_p - \tfrac{\beta}{2}F(q)\cdot p, \qquad \mathcal{S}_{\text{pert}} = \tfrac{\beta}{2}F(q)\cdot p. \tag{282}$$

For $f \in \mathcal{C}_{\text{Lan}}$, the fast Dirichlet form of $\mathcal{R}$ gives

$$\|(-\mathcal{R})^{1/2} f\|_{L^2(\mu)}^2 = \langle(-\mathcal{R})f, f\rangle_{L^2(\mu)} = \beta^{-1}\|\nabla_p f\|_{L^2(\mu)}^2, \tag{283}$$

and by density the same identity extends to $f \in \operatorname{dom}((-\mathcal{R})^{1/2})$. Since F is bounded,

$$\|F\cdot\nabla_p f\|_{L^2(\mu)} \le \|F\|_{L^\infty}\|\nabla_p f\|_{L^2(\mu)} \le \sqrt{\beta}\,\|F\|_{L^\infty}\|(-\mathcal{R})^{1/2} f\|_{L^2(\mu)}. \tag{284}$$

For the multiplier part, the Gaussian integration-by-parts estimate in momentum gives

$$\||p|f\|_{L^2(\mu)} \le \sqrt{2d/\beta}\,\|f\|_{L^2(\mu)} + (2/\beta)\|\nabla_p f\|_{L^2(\mu)}. \tag{285}$$

Therefore

$$\|F\cdot p\, f\|_{L^2(\mu)} \le \|F\|_{L^\infty}\Big(\sqrt{2d/\beta}\,\|f\|_{L^2(\mu)} + \tfrac{2}{\beta}\|\nabla_p f\|_{L^2(\mu)}\Big). \tag{286}$$

Combining the last two bounds shows that Assumption 4.1.4 holds with constants (a_1, b_1, a_2, b_2) depending only on $(\beta, d, \|F\|_{L^\infty})$ and independent of m.

Appendix G. Proof of Proposition 6.4

All operators are bounded on the finite-dimensional space $\mathcal{H}$.

(1) Assumption 4.1.1 holds for $\mathcal{R}$ by construction, with $\lambda_R = 4\Gamma$.
(2) Assumption 4.1.2 holds since both $\mathcal{V}_{\text{skew}}$ and $\mathcal{V}_{\text{pert}}$ map $\mathcal{H}_S$ into $\mathcal{H}_F$.
(3) Assumption 4.1.3 and Assumption 4.2 are both verified explicitly. The key improvement over a single-channel coupling (e.g., $H_{\text{int}} = g\sigma_z\otimes\sigma_z$) is that the double commutator $\operatorname{ad}_{\sigma_a}^2$ has kernel $\operatorname{span}\{I, \sigma_a\}$; with two channels,

$$\ker(Q_O)\cap\mathcal{H}_0 = \ker(\operatorname{ad}_{\sigma_x}^2)\cap\ker(\operatorname{ad}_{\sigma_z}^2)\cap\{\operatorname{tr}=0\} = \{0\}, \tag{287}$$

so Assumption 4.2 holds with $\ker(Q_O) = \mathbb{C}\mathbf{1}$ and $\lambda(Q_O) = 4g^2/\kappa = g^2/\Gamma$; the numerical values in Table 2 simply reproduce this exact formula.

For the flux regularity bound (23), define the bounded operator

$$\mathcal{K} := P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)P_F\mathcal{V}_{\text{skew}}P_S : \mathcal{H}_S \to \mathcal{H}_F. \tag{288}$$

Because $P_S\mathcal{V}_{\text{skew}}P_S = 0$, both $\mathcal{K}$ and $-Q_O$ annihilate the stationary direction $\mathbb{C}\mathbf{1}$. On the finite-dimensional mean-zero slow space $\mathcal{H}_{S,0}$, the kernel computation above shows that $-Q_O|_{\mathcal{H}_{S,0}}$ is positive definite and therefore invertible. We may thus set

$$\mathcal{T} := \mathcal{K}\,(-Q_O|_{\mathcal{H}_{S,0}})^{-1} : \mathcal{H}_{S,0} \to \mathcal{H}_F, \qquad \kappa_{\text{ell}} := \|\mathcal{T}\| < \infty. \tag{289}$$

Writing any slow vector as $\psi = c\mathbf{1} + \psi_0$ with $\psi_0 \in \mathcal{H}_{S,0}$ and using $g_{\mathbf{1}} = 0$, we obtain

$$P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\psi = \mathcal{K}\psi_0 = \mathcal{T}(-Q_O)\psi_0, \tag{290}$$

and hence

$$\|P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\psi\| \le \kappa_{\text{ell}}\|(-Q_O)\psi_0\| = \kappa_{\text{ell}}\|(-Q_O)\psi\|. \tag{291}$$

Thus (23) holds analytically in this finite-dimensional model, and one may take $M_R := \|\mathcal{R}\|$.

(4) For Assumption 4.1.4, decompose $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ by $\mathcal{K}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} - \mathcal{V}_{\text{pert}}^\dagger)$ and $\mathcal{S}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}^\dagger)$. Since $\mathcal{V}_{\text{pert}}$ is bounded, the energy bounds (26)–(27) hold with $a_1 = a_2 = 0$ and $b_1 = \|\mathcal{K}_{\text{pert}}\|$, $b_2 = \|\mathcal{S}_{\text{pert}}\|$.

□

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